# claude-windows / 9f6357d8-0e35-491a-a830-052e094c7f37

## Metadata

- Source: `claude-windows`
- Kind: `claude`
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\9f6357d8-0e35-491a-a830-052e094c7f37\subagents\workflows\wf_b6ef6275-de5\agent-a163be781b327f202.jsonl`
- SHA-256: `51c7399fcf6f0ed6379688f2134acefa8181840b12e3b89a543905f8f532d09f`
- Source modified: `2026-06-21T13:58:40+00:00`
- Imported at: `2026-07-05T16:48:25+00:00`
- project: `wf_b6ef6275-de5`
- session_id: `9f6357d8-0e35-491a-a830-052e094c7f37`

## Transcript

### 1. user (2026-06-21T13:56:13.272Z)

```
Read the CROSS-CUTTING REGISTERS in the engine repo (C:/Users/avidu/Projects/badminton-highlight-indexer) and khelsutra (C:/Users/avidu/Projects/khelsutra) to map what is active vs archived and the overall next-step pointers. Read engine docs/DOC_STATUS.md (the LIVING doc-lifecycle register ? which docs are DRAFT/IN-PROGRESS/DONE/archived, and any cross-repo deferred rows), engine docs/README.md (doc index) + docs/NEXT_STEPS.md, docs/CODE_MAP.md (for current architecture), and khelsutra docs/README.md + NEXT_STEPS.md. Summarize the cross-cutting next steps + which tracks DOC_STATUS marks as active/owner-gated. Flag any doc DOC_STATUS lists that contradicts reality. Return structured JSON (track="cross-cutting").
```

### 2. assistant (2026-06-21T13:56:16.018Z)

```
I'll start by reading the cross-cutting registers in both repos. Let me pull both repos first and read the key docs in parallel.
```

### 3. user (2026-06-21T13:56:21.466Z)

```
fatal: Cannot fast-forward to multiple branches.
---KHELSUTRA---
fatal: Cannot fast-forward to multiple branches.
```

### 4. user (2026-06-21T13:56:21.492Z)

```
1	# Documentation Status & Archival Register
2
3	> **Status:** ? **LIVING** ? the single board for doc **lifecycle + archival due-diligence**. **Last full review:** 2026-06-21.
4	> **Update cadence:** whenever a doc changes lifecycle state (created / shipped a milestone / delivered / archived), and re-review at each clean-slate.
5	> **Why this exists:** so we can (a) **review completeness** at a glance, (b) **update it as work finishes** to prove we're done, and (c) **never archive a foundational doc without honestly updating it first** (owner directive, 2026-06-21).
6
7	---
8
9	## 0. How to use this
10	- **Reviewing now?** Read §4 (the open findings/actions) ? that's what needs attention. §3 is the full register.
11	- **Marking something complete?** Update the doc's own **§7 progress tracker** (the source of truth for *its* completeness), then flip its row here, then ? if terminal ? run the §2 archival checklist.
12	- **This register does NOT duplicate completeness detail** ? it indexes lifecycle + archive-readiness and points to each tracked doc's §7.
13
14	## 1. Lifecycle legend
```

`DRAFT ? IN PROGRESS ? DONE/DELIVERED ? ARCHIVED`. Plus: **LIVING** (permanent reference, never "done"), **PROPOSAL/DESIGN** (owner-gated, not started), **REPORT** (a terminal record of a run/launch), **CONVENTION** (a standing rule/guide).

## 2. Archival due-diligence checklist  ? *the repeatable process (do NOT just `git mv`)*
Before a foundational/tracked doc moves to `docs/archives/`:
1. **VERIFY claims against the code** ? no stale assertions; tag load-bearing facts `[verified]`. (A doc that says "nothing built" when code shipped is *not* archive-ready ? fix it first.)
2. **Flip Status ? DELIVERED / DONE / REJECTED + a completion note** ? what shipped, where (PRs, paths, artifacts), and what was **deferred / carried forward**.
3. **Update ALL inbound references** ? `docs/README.md` index, `CLAUDE.md`, `docs/NEXT_STEPS.md`, and any cross-doc links ? the new path + status.
4. **Keep the lineage** ? ADR forward/back-links + "supersedes / superseded-by" so the evolution stays traceable.
5. **`git mv` the whole dir/file ? `docs/archives/past_projects/`** (terminal-state ONLY; live docs stay at `docs/` top level). Relative links inside become historical (note this in the doc's header).
6. **Record the move** ? flip the row here to §3e + a line in memory `current-state`.

> **Worked example (the template):** **DECOUPLED_COMPUTE ? #270** ? verified M0?M2+M3.5 delivered on GCP, flipped ??? DELIVERED with a completion note (incl. carried-forward WASB-C3/DPA/`md5`), reindexed `docs/README` + `past_projects/README`, kept the ADR-010?014 lineage, then `git mv` to `docs/archives/past_projects/DECOUPLED_COMPUTE/`.

## 3. The register

### 3a. Tracked projects ? lifecycle docs (archive when DONE; each doc's §7 is the completeness source-of-truth)
| Doc | Status | Archive-ready? | Next action |
|---|---|---|---|
| `COMPUTE_DECOUPLED_SERVING/` | **IN PROGRESS** ? M0 approved + **M1?M4 SHIPPED** (#279/#280/#282/#283); M5+ default-OFF code batch authorized (2026-06-21) | No | M5 truly-local ? M6 Cloud Run ? M8 ledger ? M9-auth ? M11 flywheel ? M12 gdrive-api (owner residual: counsel/prices/OAuth/real-run within the D17 cap) |
| `RALLY_QUALITY_RESEARCH.md` | Foundation, **tracked & ACTIVE** (§7) | No | Drives the R-series; complete when §7.1 single quality number is attributed |
| `R2_EVAL_METRIC_DESIGN.md` | IMPLEMENTED (augmenting) | No (gates R4 + dual-eval; ablation pending) | Promote-or-reject tolF1 as the gate (ablation) ? then archive with the R-cluster |
| `R1_MILESTONE_LAUNCH_REPORT.md` | LAUNCHED (#204) ? terminal record | Soon (with the R-cluster) | Keep with R-series for narrative; archive when the track wraps |
| `RALLY_DETECTION_GAP_CLOSURE.md` | DRAFT (owner-gated) | No | G1?G3 levers; part of the R/quality cluster |
| `PER_VIDEO_WORKSPACE.md` | DRAFT, IN PROGRESS (P0 = #264) | No | ? supersede-reconcile (finding #1) DONE; P3 de-hardcode base ? via #282; remaining P1?P6 (P1 v6 migration coordinates with M8's v6 cost-ledger ? version bump authored once) |
| `MULTI_SIGNAL_FUSION_PLAN.md` | P1 DONE; P2+ owner-gated | No | P3/P4 (learned fusion, audio) ? needs GPU golden features |
| `EXPERIMENT_HARNESS.md` | P1 IMPLEMENTED | No | A/B + promotion gate; supports the quality track |
| `PROMINENT_COURT_DETECTION.md` | DRAFT (owner-gated) | No | C-series; multi-track selection design |
| `EXECUTION_POLICY.md` | P1 SHIPPED; P3b next | No | Async job policy; reused by Cloud Run dispatch |
| `GOLDEN_REGRESSION_FIXTURES.md` | IN PROGRESS (owner-gated) | No | Golden eval fixtures |
| **Data layer** ? `docs/data_pipeline/` (`GOLDEN_SET_IMPLEMENTATION` · `GOLDEN_SET_PHASE1_VIDEOS` · `GOLDEN_VIDEOS` · `VIDEO_RECORDING_GUIDELINES` · `GOLDEN_DATA_SHARING`) | LIVING (data, **no-ML**) | n/a | Relocated 2026-06-21 to the isolated data pillar; tracked via `data_pipeline/README.md`. Roora *vendor* docs archived ? `archives/roora-vendor/`. |
| `DESKTOP_APP_PLAN.md` | DRAFT (owner-gated, nothing built) | No | P0 compute-feasibility spike; relates to truly-local profile |

| `OWNED_MODEL_IMPLEMENTATION_PLAN.md` | PLAN/roadmap (owner-gated) | No (authoritative roadmap) | Gen-0?Gen-2; the owned-model track |
| `PERSONALIZATION_PLAN.md` | DESIGN ? owner-adopted; Phase-0 | No | Per-user models (DB v5 store landed) |
| `UI_ALPHA_EXECUTION_PLAN.md` | ACTIVE | No | khelsutra UI alpha track |
| `I18N_PLAN.md` | Design (P0); **in implementation** (adapted) | No | Executed in `khelsutra/docs/FRIENDLY_AND_MULTILINGUAL.md` (P1?site/, P3?web/); P2 backend stays here, owner-gated |
| `POST_166_...RISK_ASSESSMENT.md` | Plan for owner review | No | Milestones/risk doc |

### 3b. Foundational / strategy references (LIVING ? archive only if **superseded**, not when "done")

| Doc | Status | Note |
|---|---|---|
| `PLATFORM_ARCHITECTURE.md` | Strategy/research | The `ComputeBackend`/`StorageBackend` registries ? realized by COMPUTE_DECOUPLED_SERVING |
| `COMMERCIALIZATION.md` | ? LIVING | Strategy; carries WASB-C3 |
| `VIDEO_LOCALITY_MODEL.md` | THINKING DOC | L0 fully-local = the truly-local profile |
| `HOSTING_PLAN.md` | PROPOSAL | **? partly overtaken by ADR-013 + the shipped site ? verify (finding #5)** |
| `UI_PROPOSAL.md` | PROPOSAL (partly shipped) | UX half **in implementation** ? `khelsutra/docs/FRIENDLY_AND_MULTILINGUAL.md`; currency banner added (finding #5 ? for this doc) |
| `STORAGE_SHARING_MODEL.md` | DRAFT (co-design pending) | ? |
| `ANALYZERS_AND_RECONCILERS.md` | DESIGN (owner-gated) | ? |
| `ROADMAP.md` / `CODE_MAP.md` / `HOW_RALLY_DETECTION_WORKS.md` / `KHELSUTRA_PRODUCT_OVERVIEW.md` | LIVING references | Keep current; not archive targets |

### 3c. Reports & terminal records (archive candidates once their track wraps)

| Doc | Status | Note |
|---|---|---|
| `REAL_FOOTAGE_VALIDATION_REPORT.md` | validation complete | Motivates R2; archive with the R-cluster |
| `RALLY_DETECTION_QUALITY_REPORT.md` | report | Quality snapshot; part of R-cluster |
| `QUALITY_ITERATIONS.md` | LIVING empirical log | **Keep** ? the paper-backbone ablation log ([[paper-coauthor-aim]]) |
| `CODE_AUDIT_AND_TEST_HARDENING.md` · `HARDENING_LOOP_HANDOFF.md` · `DEFERRED_HARDENING_PLAN.md` | COMPLETED ? **top-level pointer stubs** (archived copies exist) | **? confirm intentional stubs (per #165) vs finish-archiving (finding #4)** |

### 3d. Conventions / guidelines / templates / runbooks (permanent ? do not archive)

`PROJECT_DOC_TEMPLATE.md` · `NEXT_STEPS.md` (? refreshed 2026-06-21) · `BACKLOG.md` · `EVALUATION.md` · `ABLATION_LAYER.md` · `SETUP_NEW_MACHINE.md` · `TRACKNET_WSL_SETUP.md` · `README.md`. *(2026-06-21 relocation: the data-layer guides moved ? `data_pipeline/`; the field-ops + Roora docs ? `archives/{field-pmf,roora-vendor}/`.)*

### 3e. Already archived (`docs/archives/`)

`past_projects/DECOUPLED_COMPUTE/` (? 2026-06-21, #270) · `past_projects/low-regret-moves/` · **`field-pmf/`** (field-ops + customer-feedback, 2026-06-21) · **`roora-vendor/`** (annotation-vendor engagement, 2026-06-21) · `PER_VIDEO_OUTPUT_LAYOUT-SUPERSEDED-2026-06-21.md` · `CODE_AUDIT_AND_TEST_HARDENING-COMPLETED-2026-06-14.md` · `DEFERRED_HARDENING_PLAN-COMPLETED-2026-06-14.md` · `HARDENING_LOOP_HANDOFF-COMPLETED-2026-06-14.md` · `COMMERCIAL_READINESS_REVIEW.md` · `MONETIZATION_AUDIT.md` · `TEST_RUN_SUMMARY.md` · `research/` · `checkpoints/` · `quality-iterations/`.

### 3f. ADR log ? `docs/archives/decisions/DECISIONS.md`

**ADR-001 ? ADR-014** (14, all ratified). The compute/serving lineage: **ADR-005 ? 008 ? 009 ? 010 ? 011 ? 012 ? 013 ? 014**. ADR-014 was **RATIFIED 2026-06-21** (owner M0 sign-off for COMPUTE_DECOUPLED_SERVING).

## 4. Open due-diligence findings / actions (2026-06-21 review)

> **? Doc-restructuring done (2026-06-21, owner-directed):** field-ops/customer-feedback ? `archives/field-pmf/`, Roora annotation-vendor ? `archives/roora-vendor/`, and a no-ML **data layer** carved into `docs/data_pipeline/`. Inbound + outbound links rewired; three dir READMEs

added (the `data_pipeline` one sets the "no segmentation/stitching" boundary).
84      1. **? RESOLVED (2026-06-21) ? `PER_VIDEO_WORKSPACE.md` supersedes `PER_VIDEO_OUTPUT_LAYOUT.md`.** Folded forward the `video_slug` naming option (workspace §8.5) + the `output/GX010125_run/` manual-prototype precedent; predecessor archived ? `docs/archives/PER_VIDEO_OUTPUT_LAYOUT-SUPERSEDED-2026-06-21.md`; `CODE_MAP.md` repointed.
85      2. **? RESOLVED (2026-06-21) ? `NEXT_STEPS.md` refreshed** with a current 2026-06-21 state + a **cross-track prioritized pending list**; the 2026-06-13 owned-model/quality body is kept (clearly marked) as that track's detail.
86      3. **R-series / quality cluster is ACTIVE** (`R1`,`R2`,`RALLY_DETECTION_GAP_CLOSURE`,`RALLY_QUALITY_RESEARCH`,`QUALITY_ITERATIONS`,`REAL_FOOTAGE_VALIDATION_REPORT`,`RALLY_DETECTION_QUALITY_REPORT`). **Do NOT archive yet** ? archive as a *cluster* once R2 is promoted-or-rejected as the gate + §7.1's single number is attributed. (Owner already confirmed "leave as is", 2026-06-21.)
87      4. **Confirm the COMPLETED top-level stubs** (`CODE_AUDIT_AND_TEST_HARDENING`, `HARDENING_LOOP_HANDOFF`, `DEFERRED_HARDENING_PLAN`) are intentional pointer-stubs (per #165) ? or finish-archive them so there's one copy.
88      5. **`HOSTING_PLAN.md` / `UI_PROPOSAL.md` are PROPOSAL but partly overtaken** by ADR-013 (DO dropped) + the shipped khelsutra site. **? `UI_PROPOSAL.md` banner added (2026-06-21)** ? its UX half is now tracked in `khelsutra/docs/FRIENDLY_AND_MULTILINGUAL.md`; the `I18N_PLAN.md` localization track is likewise in implementation there. **`HOSTING_PLAN.md` banner still pending** (ADR-013 already noted in it).
89      6. **? RESOLVED (2026-06-21) ? `COMPUTE_DECOUPLED_SERVING` M0 approved + M1?M4 SHIPPED** (#279/#280/#282/#283); all §8 gating Qs locked (D7?D17); M5+ default-OFF code batch authorized (owner residual: counsel ToS/DPA, real GCP prices, OAuth app, real cloud run within the D17 $100/?10k cap).
90
91      ## 5. Maintenance
92      Update a row when its doc changes lifecycle; run §2 before any archive; re-review §4 at each clean-slate. This register + the §2 checklist are the operational form of the `CLAUDE.md` archive policy and [[working-agreement-tracked-project-docs]].
93

### 5. user (2026-06-21T13:56:21.500Z)

1       # Next steps (live) ? refreshed 2026-06-21
2
3       > **How to read:** this top section is the **current cross-track state + prioritized pending list** (refreshed 2026-06-21, replacing the stale 2026-06-13 header). The detailed **owned-model & rally-quality track** plan (2026-06-13) is **preserved below, still valid** ? it is now *one of several* active tracks. Live blow-by-blow: memory `current-state.md`. Doc lifecycle/hygiene: `docs/DOC_STATUS.md`.
4
5       ## ? Where we are (2026-06-21)
6       **DECOUPLED_COMPUTE delivered on GCP + archived** (#270); **compute = GCP only** (ADR-013, DO dropped). **`COMPUTE_DECOUPLED_SERVING`** subproject spun up (#272, **ADR-014**) ? one pure core + local/cloud profiles; **owner M0 design sign-off pending**. **Per-video workspace P0** shipped (#264); **doc-status register + archival due-diligence** in place (#273). The **khelsutra site + per-rally UI** ship in parallel (sibling track). The **rally-quality / owned-model** track (below) is the core ML work, gated on **growing the golden corpus**.
7
8       ## ? Prioritized pending ? all tracks
9       **P0 ? highest leverage / unblocks the most**
10      1. ? **`COMPUTE_DECOUPLED_SERVING` M0 approved + M1?M4 SHIPPED (#279/#280/#282/#283) + M5+ batch authorized** (owner 2026-06-21, ADR-014 ratified). All §8 gating Qs locked (D7?D17). **Owner residual** (not blocking Claude's default-OFF code): counsel ToS/DPA (M9-consent live + M11 live), real GCP prices+FX (M8 calibrate), Google OAuth app (M12 real), CF team-domain/AUD + ingress-lock; real cloud verification authorized within a **$100/?10k cap** (D17).
11      2. **[owner, ongoing] Grow the golden corpus** ? multi-court badminton + ?3 matches/sport (TT first). The single highest-leverage move; the quality track *and* full-corpus training feed on it.
12      3. **[owner + Claude · GPU] Full-corpus training (n?5)** ? a real `weights/<ver>/` (the `0.1.0-l4` bundle is n=2 plumbing-only).
13
14      **P1 ? near-term, mostly Claude-doable (default-OFF, green-gated)**
15      4. **[Claude] `COMPUTE_DECOUPLED_SERVING` M5?M12 default-OFF code batch** (owner-

authorized 2026-06-21, merge-each-on-green): M5 truly-local preset+startup-checks+L4 parity test ? M6 `backend/compute/` Cloud Run dispatch + Dockerfile (mocked) ? M8 v6 ledger+pricebook+/cost ? M9-auth Cf-Access JWT+tenant-scoping ? M11 flywheel plumbing ? M12 `gdrive-api` backend. *(M1?M4 config/input/output/device layer = ? SHIPPED.)*

5. **[Claude] Per-video workspace P1?P6** (P1 v6 DB migration ? P2 worker ? P3 per-video routing of `detector/`+`tmp/` *[the `output/chunks_*` de-hardcode base = ? done via #282]* ? P4 ? P5 Launch-Report flip ? P6 GPU parity). ? P1's v6 migration coordinates with the M8 v6 cost-ledger migration ? author the version bump once.
6. **[Claude] Ops-Agent codification into `deploy/gcp`** (every VM auto-enables observability ? [[gcp-ops-agent-enable]]).
7. **[owner] Set the ship-gate target (#172)** ? needs corpus growth.
8. **[owner-side GPU] R-series: promote-or-reject `R2` tolF1 as the gate** (the ablation) ? needs GPU-extracted golden features.

**P2 ? later / gated / deferred**
9. Rally-quality **P3/P4** (person-fusion LOVO, audio cue) ? needs GPU golden features.
10. **M0.4 scorer swap** ? TemporalMaxer (per the 2026-06-17 external review).
11. **Multisport** (table-tennis first, then pickleball).
12. Serving **M8/M9** ? per-video cost ledger, edge auth, DPA/consent (within `COMPUTE_DECOUPLED_SERVING`; owner/legal-gated).
13. **WASB-C3** weight provenance (gates *sharing* a trained model) ? legal.
14. **Rename repo off "badminton"** (owner, BACKLOG).
15. Owner housekeeping: retire the shared **DO droplet** + **DO token**; collector **`md5`** export (collector repo).
16. **PMF field validation** (owner-side, business track).

**Doc-hygiene (tracked in `docs/DOC_STATUS.md` §4):** ? #1 per-video reconcile + ? #2 this refresh; **pending:** field/PMF/Roora ? archives + a data-layer `docs/` dir (owner boundary confirm); confirm the COMPLETED top-level stubs; verify `HOSTING_PLAN`/`UI_PROPOSAL` currency.

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## Owned-model & rally-quality track ? detail (2026-06-13 plan; still valid)

**Updated:** 2026-06-13 (**rally-detection quality track**: fusion + experiment-harness design docs merged, owner-gated ? see the dedicated section below, the desktop-agent pick-up). Prior: 2026-06-10 (audit?M0?Gemini-battery session; later same day the **UI-alpha track opened**: ADR-009 ratified ? KhelSutra Guru / khelsutra.guru, private `khelsutra` repo, local-first delivery ? see `docs/UI_ALPHA_EXECUTION_PLAN.md`; **PR-1 async jobs (#16) MERGED 2026-06-11, PR #87**; **PR-2 upload/download MERGED 2026-06-11, PR #99** ? live-E2E-verified (3-part chunked upload MD5 byte-identity + compile?browser-download); next: PR-3 identity. Same day: the **8?9 quality sweep #90?#98 landed** ? ruff/mypy/ coverage-floor CI lanes, SQLite conn-close fix, config unknown-key warning, hybrid-twin collapse into `trajectory_hybrid.py`, WSL tilde+abspath live-path fixes; suite 388 green; LOVO 0.626 re-verified exact).
**Authoritative roadmap:** `docs/OWNED_MODEL_IMPLEMENTATION_PLAN.md` (+ ADR-008 for topology). **Latest blow-by-blow:** memory `current-state.md` / `session-handoff.md`. This file = the prioritized "what to do next," refreshed each session.

## ? Next-session / GPU-box pickup (2026-06-15)
**M0 in-container quick wins are COMPLETE:** ship-gate provisional blocker (**#170**), train/eval split guard (**#171**), the **doc-consistency sweep** (**#174** ? corrected 16 drifted docs vs the code), and the **P3 person-feature fusion litmus machinery is already present + suite-green** (`backend/eval/fusion_compare.py` + `ablation.py`: person-driven ?F1 / bootstrap CI / paired sign test; the real-golden `skipif` litmus waits only on GPU-extracted data). ? CI is red repo-wide due to a **GitHub Actions minutes/billing outage** (account-level, not code) ? work is local-gated + `--admin`-merged until Actions is restored (Settings ? Billing ? Actions minutes / spending limit).

**Immediate next step ? the real P3 LOVO measurement** (this box has a GPU + `torch+cu124` + `cv2`; videos and trajectory CSVs are present, but the GPU-extracted golden features + `.rallies.csv` labels are not yet here):
1. **[owner / GPU box]** `python -m backend.eval.fusion_golden --manifest output/human_lovo_manifest.json --out-dir "%USERPROFILE%\OneDrive\rally-annotator\golden-data\features\2026-06-15-n6"` ? produces `*_golden_features.json` for the 6 golden videos. (GPU + WSL + videos + trajectory CSVs + `.rallies.csv` labels are owner-side only.)
2. Artifacts land in the **shared cloud folder** `C:\Users\avidu\OneDrive\rally-annotator\golden-data\` (OneDrive auto-syncs both ways). Convention + workflow: **[`GOLDEN_DATA_SHARING.md`](data_pipeline/GOLDEN_DATA_SHARING.md)**; tracked in **#175**.
3. **[CPU dev/agent session]** `python -m backend.eval.fusion_compare --features-dir <shared>\features\2026-06-15-n6 --record` + `python -m backend.eval.ablation --features-dir <shared>\... --granularity group` ? real person-feature ?F1; copy the JSONs into `output/` to also pass the `skipif` litmus tests (`tests/test_ablation.py::test_litmus_reproduces_gate_a`).

**Also queued (owner-gated / pending input):**
- Ship-gate target calibration ? **#172** (owner decision; needs corpus growth).
- Tests reorganization ? blocked on the owner's grouping metric (provide it ? an agent will do it).
- Restore CI ? the Actions outage is account-level (billing/minutes), not a repo/config bug.

**Leaving (bulky / M2 ? no start without owner go):** ActionFormer (M0.4), event-index DB migration (M0.1), Gemini-silver purge (M0.3), cost-to-Gen-2 benchmark + ByteTrack deprecation, multisport, repo rename, PMF execution.

## Where we are (one paragraph)
ADR-008 ratified ("repo split driven by productionization, not isolation"): training lives in-repo (`training/`)
behind a CI-enforced import wall; the featurizer is single-sourced (`backend/features/rally_seq.py`, train==serve);
the **Gen-0 harness ships** (`python -m training.gen0.harness --manifest output/human_lovo_manifest.json --floor
eval_baselines/heuristic_lovo_n6.json --version <semver>`) and produced artifact v0.1.0 (LOVO **0.626**, floor passed,
sign test 5/6 wins p=0.109 ? honestly not significant, ship=False). The **n=6 owned corpus is in the collector**
(262 rallies, Proprietary, commercial export = owned data only after the ShuttleSet reclassification). The **Gemini
battery is done** (see table) ? the moat is quantified.

## The quality table that now drives strategy (IoU 0.5 vs human golden labels)
| Path | Mahadevpura (clean-ish) | Kushagra (multi-court) |
|---|---|---|
| Heuristic (velocity windowing) | 0.657 | 0.157 |
| **Gen-0 owned head** (LOVO, n=6) | 0.744 | 0.561 |
| Two-stage WASB+Gemini refine | 0.83 | ? |
| **Gemini wrapper** (`gemini-flash-latest`) | **0.93** | **0.66** |

**Reading:** clean footage is commoditized (serve it with Gemini for cents). **Multi-court drops the commodity
wrapper to 0.66 ? that is the owned model's ship target** (its FPs are background-game pickups + dead-time merges,
the exact contrast-metric confound). Gen-0 is 0.10 behind with only n=6 ? a corpus-growth-sized gap, not an
architecture-sized one. Two-stage refine = the cost-efficient serving path for LONG tapes (uploads candidate clips
only); wrapper 503-flakiness (observed all session) makes the provider-fallback registry load-bearing.
Baselines tracked in `eval_baselines/` (heuristic_lovo_n6.json, gemini_wrapper_2026-06-10.json).

## Rally-detection quality track (design merged 2026-06-13, owner-gated; implementation in progress)

89 **This is the track a desktop/local agent is slated to pick up.** Design docs landed (PR #112 + session). All implementation is **owner-gated** (per explicit in the plan); each step = ONE PR off `master`.
90
91 **Hardening loop (T1-T5 positive tests + audit + finale) COMPLETE (2026-06-14, PRs #156/#157/#160/#161/#163 + #165 + final records).** See archived `docs/archives/CODE_AUDIT_AND_TEST_HARDENING-COMPLETED-2026-06-14.md` (Living Log with every pre/post/review) and `docs/archives/HARDENING_LOOP_HANDOFF-COMPLETED-2026-06-14.md` (full rules + STATE). Top-level now pointer stubs per #165 review. All 5 items + verification + archive done; clean slate.
92
93 - **Docs:** `docs/MULTI_SIGNAL_FUSION_PLAN.md` (shuttle + person + audio, one fusion layer) ·
94 `docs/EXPERIMENT_HARNESS.md` (A/B + shadow + promotion gate, zero-regression) ·
95 `docs/ROORA_LABELING_GUIDELINES.md` (v8) · `docs/HOW_RALLY_DETECTION_WORKS.md` (2-layer mental model).
96 - **Key finding (no re-investigation needed):** the owner's "held-shuttle counted as a rally" bug is **NOT a missing capability** ? `backend/pipeline/detectors/rally_gate.py` already implements the net-crossing **"Gate A"** (`apply_gate(..., min_crossings=2)`) that kills service-feed/held-shuttle windows, but it is **only called from eval drivers** (and there was no `min_crossings` knob in `IndexingConfig`). **Wiring Gate A into the live segmenter (now via fusion_hybrid) was the single cheapest, highest-confidence quality win.** (P2 FusionSegmenter specifics: registry `fusion_hybrid`, **default-OFF**, seeds from substrate, persists `net_crossings` + optional person features, optional served Gate A/B via config knobs; adversarially reviewed; suite green. See `tests/test_served_gate_a_regression.py` for the honest finding that on production windowing Gate A is neutral/negative in some cases, hence kept opt-in with knob for safety, and the leaderboard revert note.)
97 - **Current status (2026-06-14):** P1 (person cue extraction to reusable `PersonDetector`) + P2 (FusionSegmenter + Gate A signal + served handoff gating) + experiment harness P1 DONE (adversarial review + NO-REGRESSION; suite green). **P3 next (this PR + follow-ups):** learned fusion (person features ? harness `compare` LOVO lift on the 6 golden videos; eager-person-compute optimisation); then P4 audio via hit_detector.
98 - **Pick-up order (each = ONE PR off `master`, owner approval first, flag-gated default-OFF, promote only on measured LOVO):**
99 1. (P2 first-step, largely done via fusion_hybrid) Wire Gate A into production served path + config knob (min_crossings); apply in hybrids at runtime. (Now available opt-in; served neutral on production windowing per litmus ? kept opt-in with knob for safety.)
100 2. P3 first step (this work item): harness `compare` LOVO litmus for person-on variant (players_in_court + two_sided + colocation etc. from `fusion_features`) on the 6 golden; add the eager compute optimisation note/landing. Pure + harness (testable here).
101 3. P3/P4 continuation: audio cue integration + full learned promotion if lift proven.
102 - **Discipline (from EXPERIMENT_HARNESS.md):** every new cue **flag-gated, default OFF**; promote to default **only on measured LOVO lift** through `run_regression.py` (exit 0/1/3); pinned `CONTROL` ? rollback = config flip. Harness re-uses ~80% existing code.
103 - ? Gate A, person cue logic, harness, and pure fusion paths are testable here; the heavier real-video / golden LOVO / ablation confirmation waits on the GPU-extracted golden features + `.rallies.csv` labels, which are not yet present in this checkout.
104 - **External due-diligence review inputs (2026-06-17) ? PICK UP AFTER current ideas are exhausted; full details + citations in [`RALLY_QUALITY_RESEARCH.md`](RALLY_QUALITY_RESEARCH.md) §5.6.** An owner-commissioned 31-reference review of the quality report *validated* the methodology and surfaced these **future** levers (research inputs, not active work, not new claims; verify citations before any submission):
105 1. **"First-mile" paper framing (highest-value):** stroke-classifiers (BST/DSTA) + the big datasets (ShuttleSet ~36k strokes, Fine-Badminton ~27k actions) all **assume a pre-trimmed rally clip already exists** ? we solve the ignored prerequisite. Pitch the paper as *"the missing untrimmed-video preprocessing pipeline that lets BST/MonoTrack run on raw footage."*
106 2. **Ground R2 in Action-Spotting / Precise-Event-Spotting** (SoccerNet/TenniSet/T-DEED) ? turns R2 into a defensible publishable contribution.
107 3. **Add domain baselines for any submission:** MonoTrack + BST on ShuttleSet / Fine-Badminton (we have only heuristic + Gemini today).
108 4. **Prefer TemporalMaxer over ActionFormer** for the M0.4 scorer swap (parameter-free max-pool, ~2.8× fewer GMACs, small-N robust) ? see "M0.4 next increment" below.
109 5. **Court polygons ? pose ? the rally-END lever:** masking on-court players can resolve *occluded end-of-rally states* (the #1 remaining gap vs Gemini), not just spectator noise.

6. **Name the eval?serve lesson "pipeline distribution skew"** ? a citable paper "lesson."

**Discipline reminder:** owner approval before starting any owner-gated item; one PR per step; babysit the PR (poll/fix/reply) until merge observed, *then* advance.

## Prioritized next steps
1. **[owner, in progress] GROW THE GOLDEN CORPUS** ? the single highest-leverage move; every lever below feeds on it.
   Priority order for new videos: **(a) multi-court badminton** (the target distribution AND where the ship target
   lives), **(b) ?3 matches per new sport ? table tennis first** (WASB transfers at F1 0.909; pickleball labels are
   corpus-gold but not trainable until a clean MIT/Apache ball detector exists), (c) capture variety per
   `VIDEO_RECORDING_GUIDELINES.md`; **adult sessions only for the training pool** (consent guardrail). Per video:
   label ? `curate import-local --normalize --license Proprietary --fps <true fps>` ? re-run the Gen-0 harness.
   The paired sign test needs n: at n=10 with 9 wins, p?0.011 ? significance is reachable this month.
2. **[owner decision] Set the ship-gate target** ? now informed: floor = heuristic 0.480 (necessary), **multi-court
   reference = wrapper 0.66** (the number that makes the owned model worth serving), clean-footage ceiling = 0.93
   (not the target ? Gemini serves that tier). Suggested shape: "LOVO F1@tIoU0.5 ? 0.70 on multi-court folds with
   paired-sign p<0.05 vs heuristic" ? owner to ratify/adjust.
3. **M0.4 next increment ($0/local):** swap the Gen-0 LogReg for a lightweight temporal head (MIT, minutes on the
   2070S ? NOT the heavyweight Gen-2 stack) inside the existing harness; per-video threshold calibration (the
   LOVO-vs-oracle gap is visible per fold in the harness output). **Prefer TemporalMaxer** (parameter-free
   max-pool, ~2.8× fewer GMACs, small-N robust) over ActionFormer/ASFormer per the 2026-06-17 external review
   (`RALLY_QUALITY_RESEARCH.md` §5.6) ? better fit for the cheap/local/fast mandate.
4. **Multisport thread:** ingest Roora pickleball/TT CSVs (`import-local --normalize`); merged-prototype LOVO across
   badminton+TT once TT has ?3 matches; identify a clean pickleball ball detector. ? TT-on-WASB serving stays gated
   by C3 (provenance multiplies per sport).
5. **PMF validation (cheaper than any engineering month):** C13 academy interviews (+ the two added questions:
   "multi-court footage nobody watches?" / "pay per-match to index it?"); camera pilot at the warmest 2-3 academies
   (consent at capture; demo served via the Gemini path or human-polish ? the reel pattern exists:
   `output/MahadevpuraSingles_highlights.mp4`). **Field kit COMPLETE (2026-06-12, owner ratified the
   physical-visit approach; reframed product-feedback-first per owner spec):**
   `docs/PRODUCT_FIELD_ASSOCIATE_FLYER.md` (the recruiting flyer ? owner fills ?/contact placeholders and recruits) +
   `docs/FIELD_VISIT_PLAYBOOK.md` (the hired associate's visit playbook: coach + student script, demo-after-Q6
   rule, ground rules, Phase-2 record-and-show-back stretch goal) + `docs/FIELD_VISIT_ANSWER_SHEET.md` (per-visit
   capture form incl. demo-reaction + student blocks) + `docs/PMF_FIELD_SIGNALS.md` (G/A/R anchors + **pre-registered
   decision rules** ? PROCEED / PIVOT A-C / FALSIFIED thresholds fixed before the first visit).
6. **Housekeeping:** ~~rotate the Gemini API key~~ **DONE 2026-06-10** (owner-verified working; confirm old-key
   deletion in AI Studio); **rename repo off "badminton"** (owner action,
   in BACKLOG); collector=SoR confirmation; off-peak niceties: none pending ? the

battery completed.
147
148     ## Pointers / artifacts
149     - Roadmap: `docs/OWNED_MODEL_IMPLEMENTATION_PLAN.md` · Topology: ADR-008
(`docs/archives/decisions/DECISIONS.md`) ·
150       Study: `docs/archives/research/OWNED_MODEL_TRAINING_STUDY.md` · Quality history:
151       `docs/archives/quality-iterations/2026-06-multivideo-lovo/`.
152     - Harness: `training/gen0/harness.py` (one command ? train ? LOVO ? gate ? artifact
bundle; weights gitignored,
153       manifest reviewable). Featurizer: `backend/features/rally_seq.py` (FEATURE_SCHEMA v1).
154     - Corpus: collector DB (system-of-record, 6 matches/262 rallies) · eval manifest
`output/human_lovo_manifest.json`
155       (gitignored) · durable trajectories + Roora files: `C:\Users\avidu\Projects\Annotation
Setup\Collect\Trajectories\` ·
156       labels: `...\Collect\Golden Labelled\`.
157     - Gemini ops learnings: serial uploads (`indexing.ai_max_workers=1`) + chunk re-encode
158       (`indexing.ai_chunk_scale_width=1280`) + 6-attempt exponential generate retry ? a
cold/prepaid project burst-throttles
159       with a MISLEADING "credits depleted" 429; ramp gradually.
160     - ? No owned-model implementation beyond greenlit M0.x until owner greenlights the next
milestone. Committed next
161       PLATFORM slice = async job model (single-tenant).
162

### 6. assistant (2026-06-21T13:56:25.511Z)

The repos have a multi-branch fast-forward issue. Let me check the branch state and continue
reading the remaining docs.

### 7. user (2026-06-21T13:56:28.774Z)

=== ENGINE branch ===
feat/gcp-ops-agent-and-dryrun-guide
c3dbd00 feat(gcp): always-install Ops Agent on GPU VMs + a cloud-GPU dry-run guide
=== KHELSUTRA branch ===
master
7750afc docs(dry-run): Path-B live e2e DONE ? in-UI bucket?reel CUJ + screencast (DoD §6 box 3
?) (#57)
?? .claude/
?? site/access.db

### 8. user (2026-06-21T13:56:28.784Z)

1       - [RALLY_QUALITY_RESEARCH.md](RALLY_QUALITY_RESEARCH.md) ? **? Foundation doc** for
closing the local-vs-Gemini rally-detection gap: the measured gap + the code-grounded over-merge
diagnosis ("shuttle moving ? rally"), an inventory of the eval/experiment harness + its gaps,
the strengthened evaluation framework (segmental Edit/F1@k that make over-segmentation first-
class, golden-corpus + active learning, Gemini-agreement as a *shadow* signal), an
**adversarially-verified, cited external-research synthesis** (TAL/TAS boundary heads, racket-
sports rally-state cues, small-data, eval rigor), a **prioritized cheap?durable roadmap with
falsifiable success criteria**, and (§7) a **tracked project plan** ? work-item checklist,
sequencing, the single quality **number** attributed to the effort, and a **definition of done**
(project completes when all tiers land + the number is recorded). The plan that the
[QUALITY_ITERATIONS.md](QUALITY_ITERATIONS.md) log executes against; ties together
EXPERIMENT_HARNESS / MULTI_SIGNAL_FUSION_PLAN / ANALYZERS_AND_RECONCILERS /
OWNED_MODEL_IMPLEMENTATION_PLAN.
2       - [RALLY_DETECTION_GAP_CLOSURE.md](RALLY_DETECTION_GAP_CLOSURE.md) ? **Tracked execution
doc (2026-06-18)** for closing the remaining rally-detection accuracy gap **per video, before
launch**, driven by the growing golden corpus. First ground-truth clip (GX010141) reframed it:
boundaries are tight (R2/R3/R4 holds) ? the gap is **detection**: (G1) recall (local & Gemini
both miss ~half on hard clips, and *different* rallies ? complementary), (G2) precision (local
over-fires), (G3) **rally-END over-extension on the quick-return-after-point** (shuttle still
moving ? needs a *player-state* signal). Levers (owner-gated): player-kinematics + shuttle-in-
hand / sustained-invisibility end-rule with reverse-timeline backtrack, `rally_gate` Gate A

wiring, complementary fusion. **Peer** of `RALLY_QUALITY_RESEARCH.md` (execution tracker, not research); §7 is the source of truth.
3 - [PROMINENT_COURT_DETECTION.md](PROMINENT_COURT_DETECTION.md) ? **Design + experiment (2026-06-19, default-OFF, owner-gated flip)** for the **multicourt G2 over-fire**: identify the **prominent (foreground) court** (the one covering the majority of the frame) and keep only rallies whose shuttle lives inside it ? re-defining the shuttle signal `(x,y,visible)` ? `(x,y,visible-AND-inside-prominent-court)` *before* windowing, so a background/adjacent-court shuttle never forms a rally. 2D floor polygon (MVP) ? **pseudo-3D play-volume** (refinement, so high clears aren't clipped). Born from the GX010142 rally-#1 background-court FP; reuses `fusion_features.point_in_polygon` + `FusionConfig.court_polygon`. §8 progress tracker is the source of truth.
4 - [RALLY_DETECTION_QUALITY_REPORT.md](RALLY_DETECTION_QUALITY_REPORT.md) ? **Technical-paper checkpoint (2026-06-17)** of rally-detection quality: the honest measured state (over-segmentation milestone tolerance-F1 0.091?0.323 at n=11, p=0.002; merge-rate 0.694?0.150; first real-footage ground-truth at-par with START at Gemini parity), the task-faithful R2 metric, a detailed Gemini comparison, next steps + expected wins, areas yet to be explored, and a publishability / path-to-submission verdict. Renders to a polished PDF; the externally-shareable "toward a paper" summary of `RALLY_QUALITY_RESEARCH.md` + `QUALITY_ITERATIONS.md`.
5 - [EVALUATION.md](EVALUATION.md) ? How we measure and improve quality.
6 - [EXPERIMENT_HARNESS.md](EXPERIMENT_HARNESS.md) ? Design (P1 SHIPPED) for A/B + shadow + SxS experimentation on rally-detection variants with **zero production regression** ? experiments are decoupled from deploy (new cues default OFF, promote only on measured LOVO lift, rollback = flip a flag). Composes the existing `backend/eval/` machinery (`calibration`, `regression`, `metrics`); implemented in `backend/eval/experiment.py`.
7 - [QUALITY_ITERATIONS.md](QUALITY_ITERATIONS.md) ? **Living log** of rally-detection quality iterations (hypothesis ? what shipped ? measured/expected effect ? lesson). The reverse-chronological record; terminal-state snapshots get archived under `archives/quality-iterations/`.
8 - [ANALYZERS_AND_RECONCILERS.md](ANALYZERS_AND_RECONCILERS.md) ? Design (owner-gated, NOT started): a **signal-fusion framework** for rally detection ? producers emit confidence-scored **signals** at three temporal **scopes** (frame detectors / window analyzers / session analyzers), a learned **scorer** weights them, and an auditable **report-first reconciliation** pass lints the decision against the evidence. The spine: **no special-casing in the decision path** (signals?scorer, never `if/else`). Worked examples: a service-fault analyzer, a warm-up/practice span detector, and consuming a per-frame shuttle-state detector. A peer to `EXPERIMENT_HARNESS.md` and `MULTI_SIGNAL_FUSION_PLAN.md` (ADR-007 modular rally layer).
9 - [PERSONALIZATION_PLAN.md](PERSONALIZATION_PLAN.md) ? Design (owner-adopted **hybrid**; Phase-0 landing) for the **per-user personalization feature on a generalization-first baseline**: a thin per-user *tuning* layer gated by the honest small-N statistics (a **min-N promotion floor** + a **structural-guard exemption**, never a service gate) + a **contribution flywheel** (free-tier videos pool into the owner-trained baseline; *"a tool that gets better the more you help it"*). Records the **locked owner decisions** (identity, storage, `min_n` semantics, Gate-A default, free-tier pooling). Built so far: per-user promotion gate (#141) + held-out-USER learning curve (#142).
10
11 - [GOLDEN_REGRESSION_FIXTURES.md](GOLDEN_REGRESSION_FIXTURES.md) ? **Design (owner-gated, NOT started):** video-free, non-attributable regression fixtures so a clean clone runs the rally-seg suite with **no video/GPU/DB**. Two layers ? the post-detector freeze + **dual gates** (characterization + quality-floor) reusing the `backend/eval/` stack, and the **SEALED-FIXTURES** privacy filter (de-attribution-at-source, tiered public/key-gated, shuttle-only public tier). Carries a cited **IN/US/EU legal memo** (derived-data IP/privacy), a threat model, and a **§7 progress tracker + definition-of-done**. Extends [GOLDEN_DATA_SHARING.md](data_pipeline/GOLDEN_DATA_SHARING.md).
12
13 ## ? Documentation status & governance
14 - [DOC_STATUS.md](DOC_STATUS.md) ? **? living Documentation Status & Archival Register**: every foundational/tracked doc's lifecycle + archive-readiness, the **archival due-diligence checklist** (*honestly update ? then archive*), and the open doc-hygiene findings. Review here; update rows as docs complete.
15
16 ## Product / platform plans
17 - [COMPUTE_DECOUPLED_SERVING/](COMPUTE_DECOUPLED_SERVING/README.md) ? **? Active tracked project (2026-06-21, `DRAFT` owner-gated):** the productionization successor to DECOUPLED_COMPUTE (its deferred M3/M4). **One pure core (DETECT?WINDOW?STITCH) that never branches on deployment profile**, with a `ComputeBackend` (where it runs) + `StorageBackend`

(what bytes) selected by **one startup config**. Ships **truly-local** (local/USB + local GPU, in-process stitch, zero cloud) + **cloud-serving** (upload<2GB / gdrive / bucket ? a **Cloud Run GPU job runs detect AND stitch**, metered into a per-job cost ledger) + **cpu-mock** (CI). Includes [TESTING_STRATEGY.md](COMPUTE_DECOUPLED_SERVING/TESTING_STRATEGY.md) (profile-PARITY proof for $0), [architecture.svg](COMPUTE_DECOUPLED_SERVING/architecture.svg), a [runlogs/](COMPUTE_DECOUPLED_SERVING/runlogs/) posterity dir, and **ADR-014** ([lineage ADR-009?010?011?012?013?014](archives/decisions/DECISIONS.md)). §7 M0?M10 tracker is the source of truth.
18 - [DECOUPLED_COMPUTE/](archives/past_projects/DECOUPLED_COMPUTE/README.md) ? ? **DELIVERED + ARCHIVED (2026-06-21).** Lifted the rally pipeline onto a **rented cloud-native GPU** doing bucket-driven `train` + `infer`: storage/compute seams (PRs #246?#259) + the **M3.5 native WASB runner**; **M0?M2 + M3.5 proven on a GCP L4** (M2 ? `weights/0.1.0-l4/`). Delivered on **GCP** ? DigitalOcean dropped for compute ([ADR-013](archives/decisions/DECISIONS.md)). **M3/M4 (serving endpoint + per-video billing + scale-to-zero) deferred ? the "Cloud Run + GPU serving" subproject** ([ADR-012](archives/decisions/DECISIONS.md)). Snapshot under `docs/archives/past_projects/DECOUPLED_COMPUTE/` (incl. `DEPLOY_REPORT_GCP_2026-06-20.md`).
19 - [DESKTOP_APP_PLAN.md](DESKTOP_APP_PLAN.md) ? **Tracked scoping doc (2026-06-18, owner-gated, `[design]` ? NOTHING built):** package the local pipeline as a **cross-platform desktop app** for non-technical players ? plug in the camera's USB/SD card, click once, get back **only the highlight reels**, fully on-device (no upload, no original-video bloat). The centerpiece is the **de-WSL native-compute port** (run WASB natively per-OS: Linux/Windows CUDA · macOS MPS/CPU ? drop WSL2), and its **make-or-break unknown**: is CPU acceptable on a typical *no-GPU* enthusiast laptop (WASB is ~3.4× real-time on a laptop 2070S)? §7 tracker = **P0 compute-feasibility spike ? P1 native detector ? P2 app shell ? P3 packaging/installers**. Realizes the **L0 fully-local** tier of [VIDEO_LOCALITY_MODEL.md](VIDEO_LOCALITY_MODEL.md); is the **`LocalDesktop` ComputeBackend** of [PLATFORM_ARCHITECTURE.md](PLATFORM_ARCHITECTURE.md) §3.2; the smaller/export-clean owned model ([OWNED_MODEL_IMPLEMENTATION_PLAN.md](OWNED_MODEL_IMPLEMENTATION_PLAN.md)) is one answer to the compute crux.
20
21 ## Practical / ongoing
22 - [SETUP_NEW_MACHINE.md](SETUP_NEW_MACHINE.md) ? **? The canonical "spin up a box ? ready for training & inference" runbook.** Copy-pasteable, human-followable, **verified end-to-end on DigitalOcean (2026-06-20)** ? but **compute is now GCP-only (ADR-013)**, see [`deploy/gcp/README.md`](../deploy/gcp/README.md): credentials ? provision ? bootstrap ? suite (922 passed) ? secret-free CPU inference (no GPU/keys) ? real Gemini inference ? Gen-0 training, with pointers to the GPU/native-detector + bucket paths. The `deploy/digitalocean/*` scripts are provider-agnostic and reused on GCP.
23 - [TRACKNET_WSL_SETUP.md](TRACKNET_WSL_SETUP.md)
24 - Evaluation harness and calibration drivers (in `backend/eval/`)
25
26 ## Data layer (collection · annotation · golden sets ? **no ML**)
27 - [`data_pipeline/`](data_pipeline/README.md) ? **the isolated data pillar**: how footage is recorded (`VIDEO_RECORDING_GUIDELINES`), golden sets are created + which videos (`GOLDEN_SET_IMPLEMENTATION` · `GOLDEN_SET_PHASE1_VIDEOS` · `GOLDEN_VIDEOS`), and data is shared/ingested (`GOLDEN_DATA_SHARING`). Deliberately **no segmentation/stitching/model** detail ? that's the ML pillar (top level + `COMPUTE_DECOUPLED_SERVING/`). The **annotation-vendor (Roora)** engagement is archived under `archives/roora-vendor/`.
28
29 ## Conventions & templates
30 - [PROJECT_DOC_TEMPLATE.md](PROJECT_DOC_TEMPLATE.md) ? Reusable skeleton for a **tracked project doc**: status header · decisions-locked · progress tracker (?/?/?) · definition-of-done · archived on completion. Standardizes the emergent pattern in `RALLY_QUALITY_RESEARCH.md` §7 and `GOLDEN_REGRESSION_FIXTURES.md`. Use for any substantial design/project work.
31
32 ## Historical documents
33 Completed plans, old research, ADRs, and checkpoints live in [`archives/`](archives/).
34
35 - [CODE_AUDIT_AND_TEST_HARDENING.md](CODE_AUDIT_AND_TEST_HARDENING.md) ? **COMPLETED** (2026-06-14; T1?T5 + full loop + verification per #162). Living record of the June 2026 plumbing / contract / architectural audit, 626-test infrastructure, positive test plans for the 5 identified issues, multi-layer regression protocol (including ablation/experiment layer), and pre/post-change verification checklist. See `docs/archives/CODE_AUDIT_AND_TEST_HARDENING-COMPLETED-2026-06-14.md` and `docs/archives/checkpoints/2026-06-14-hardening-loop-complete.md` for the final archived state. (Top-level is now pointer stub per review; full content in

archives/.)

- **Field-ops / customer-feedback** (recruiting flyer · field-visit playbook · PMF) ? [`archives/field-pmf/`](archives/field-pmf/) (archived 2026-06-21; PMF validation still an active *business* thread ? see `NEXT_STEPS.md`).
- **Roora annotation-vendor** engagement (labeling brief · vendor bake-off) ? [`archives/roora-vendor/`](archives/roora-vendor/) (archived 2026-06-21).
- Superseded: per-video output layout ? [`archives/PER_VIDEO_OUTPUT_LAYOUT-SUPERSEDED-2026-06-21.md`](archives/PER_VIDEO_OUTPUT_LAYOUT-SUPERSEDED-2026-06-21.md) (? `PER_VIDEO_WORKSPACE.md`).

See `archives/README.md` for the umbrella policy and `archives/past_projects/README.md` for classification of small tech debt / starter projects.

#### 9. assistant (2026-06-21T13:56:33.850Z)

Now let me read the khelsutra cross-cutting docs and the engine CODE_MAP. Note: the engine is on a different branch (`feat/gcp-ops-agent-and-dryrun-guide`, not the `feat/serving-m4-device-context` mentioned in the snapshot), and khelsutra has uncommitted untracked files. Let me read the khelsutra registers.

#### 10. user (2026-06-21T13:56:34.356Z)

# khelsutra docs index

## Tracked work (this repo)
- **`DRY_RUN_CUJ.md`** ? **? active goal (owner directive 2026-06-21):** the UI drives the tool on a
  **GCS-bucket video** ? review/select ? compile ? **reel downloaded local**, verified in Hindi. Grounds
  the integration gap (no in-UI bucket-ingest **screen** yet ? the engine `/api/process` already accepts
  `gs://`, so Path B = SPA-only/B-P2) + a cheapest-path runbook + open decisions. `DRAFT`
  (owner-gated). Milestone of `LOCAL_APPLIANCE_ARCHITECTURE.md`.
- **`FRIENDLY_AND_MULTILINGUAL.md`** ? **? active focus (owner directive 2026-06-21):** make the
  product super user-friendly for non-tech-savvy coaches/parents **and** localize it in Hindi +
  Kannada, across both surfaces (`web/` SPA ? react-i18next, `site/` ? a shared `t()` shim). The
  implementation project for the engine's `I18N_PLAN.md` (l10n) + the UX half of `UI_PROPOSAL.md`.
  `DRAFT` (owner-gated). Archived when en+kn+hi render on both surfaces.
- **`LOCAL_APPLIANCE_ARCHITECTURE.md`** ? the single-box, UI-powered rally indexer + reel
  generator (KhelSutra Studio): the `web/` SPA drives the full ingest ? index ? reel loop on one
  box behind an edge gate; the single-box deployment profile of the engine's decoupled
  architecture. Umbrella for the console; per-rally selection is one screen of it. `DRAFT`.
- **`PER_RALLY_SELECTION.md`** ? interactive pick-rallies ? reel builder (RallyPicker). The alpha
  SPA over the engine's rally index; hybrid grid + deterministic query bar on the **existing**
  `/api/query` + `/api/compile` contract. `IN PROGRESS` (decisions locked).
- **`REQUEST_ACCESS_FORM.md`** ? segmented lead-capture form + Worker/D1 backend (shipped).
- **`PORTABILITY.md`** ? the Node+SQLite VM-host runbook (Cloudflare ? droplet parity).

## Engine planning docs
Canonical planning docs currently live in the engine repo (`badminton-highlight-indexer/docs`):
`UI_PROPOSAL.md` · `HOSTING_PLAN.md` · `UI_ALPHA_EXECUTION_PLAN.md`. They migrate here when the

27 control plane lands (per ADR-009 in the engine repo's `docs/archives/decisions/DECISIONS.md`).
28

### 11. user (2026-06-21T13:56:34.816Z)

File does not exist. Note: your current working directory is C:\Users\avidu\Projects\badminton-highlight-indexer.

### 12. assistant (2026-06-21T13:56:37.868Z)

khelsutra has no `docs/NEXT_STEPS.md`. Let me look for the equivalent and check the khelsutra docs directory, plus read the engine CODE_MAP and check for cross-cutting registers in khelsutra.

### 13. user (2026-06-21T13:56:39.007Z)

No files found

### 14. user (2026-06-21T13:56:39.308Z)

1 # CODE_MAP.md
2
3 **Purpose:** cold-start navigation map for the badminton-highlight-indexer ? a self-hosted video ? highlight-reel pipeline (CV/AI rally detection ? SQLite ? ffmpeg stitch) with a segmenter-agnostic eval/calibration stack, typed config, and per-video observability reporting.
4
5 **Regenerate:** rerun the `build-code-map` workflow (parallel subsystem readers ? this doc). Re-run after any registry/contract/WSL-interface change, or after a directory restructure (this revision follows #40).
6
7 ? **2026-06-10 audit-hardening sweep changed several behaviors documented below ? trust git history over stale lines until the next full regen:** `add_video` is now an **UPSERT** (no FK-cascade wipe on re-ingest; segmentation is destroy-AFTER-confirm via `segment_watermarks`/`prune_segments`); detector artifacts are keyed by **video_id**, not basename; **`indexing.wasb.*` is now a real typed config** (was silently dropped); `timeout_sec` defaults **1800s** (was 0/hang-forever); SQLite opens with **WAL + busy_timeout**; `match_intervals` requires real overlap (iou>0); `run_regression` baseline lives in tracked `eval_baselines/` and **exits 3 on a missing baseline**; the API validates `video_path`/`output_filename`. New tests: `test_api_endpoints`, `test_compiler`, `test_wasb_runner`, `test_reingest_safety`, `test_detector_keying`, `test_config_wasb`, `test_path_safety`, `test_reliability_hardening` (+ collector `test_licensing_gate`, `test_import_guard`, `test_cvat_robustness`).
8
9 **LEGEND**
10 - `@path` = file (always repo-relative)
11 - `::sym` = symbol (class/func/const) in the nearest preceding `@path`
12 - `->` = dataflow / call / "produces"
13 - `§` = data contract (canonical shape)
14 - `?` = gotcha / footgun
15 - `?` = registry / plugin extension point
16 - `WSL` = runs inside WSL2 conda env, NOT the Windows Python process
17
18 ---
19
20 ## 0. REPO LAYOUT ? where new files go (READ BEFORE committing a new file)
21
22 Top-level tree and the **placement rule** for each kind of file. When unsure, prefer an existing
23 **registry / extension point** (`?`) over a new top-level file.
24
25 | You're adding? | Put it in? | Notes |
26 |---|---|---|
27 | Backend / runtime code | `backend/<subsystem>/` |

`pipeline/{segmenters,detectors,validators}`, `providers/`, `sports/`, `annotations/`, `infrastructure/`, `reporting/`, `stitcher/`, `utils/`, `config/`, `features/`, `api/`. Most additions are **register a class in the relevant `?` registry**, not a new module tree. |
28 | **Eval LIBRARY** (importable, pure ? metrics, calibration, a scorer, a feature) | `backend/eval/` | Pure core, no side-effects at import; CLI/I-O lives in a separate driver. These are **imported widely** (e.g. `calibrate_wasb` by 8 sites) ? they are library, not scripts; do NOT move them out. |
29 | **Standalone run-driver / one-off entry CLI** (has `__main__`, NOT imported by app code) | **`scripts/`** | e.g. `scripts/run_phase3_eval.py`, `scripts/run_process_and_stitch.py`. Run as `python scripts/<name>.py`. If it needs `load_dotenv()`/`sys.path` before importing backend, add `"scripts/<name>.py" = ["E402"]` to `ruff.toml`. |
30 | **Ops / maintenance tool** (not part of the served app ? cache GC, batch admin) | `tools/` | Run as `python -m tools.<name>` (it's an importable package). e.g. `tools/cleanup_caches.py`. Keep the destructive decision path pure + unit-tested. |
31 | Training code (model fitting) | `training/` | Behind the CI-enforced import wall (ADR-008) ? `backend.main` must not import it. Own `requirements-train.txt`. |
32 | A test | `tests/test_*.py` | pytest auto-discovers; the full-suite lane has a **70% coverage floor**. Pure/offline + mocked (no GPU/WSL/network). |
33 | A doc (design, plan, living log) | `docs/` | Index it in `docs/README.md`. Only **terminal-state** (DONE/REJECTED) work moves to `docs/archives/`. |
34 | A throwaway repro / debug script | `scratch/` | **Gitignored** + excluded from ruff/pytest. Never imported by app code. |
35 | A model artifact (weights + manifest) | `artifacts/<name>/<ver>/` | ADR-008: **manifest committed** (provenance), **weights gitignored**. |
36 | Eval baseline / leaderboard JSON | `eval_baselines/` | Tracked (survives a clean checkout); the no-regression gate reads it. |
37 | **Committed regression fixture** (video-free golden) | `eval_baselines/fixtures/<slug>/` | Tracked, **LF-pinned** (`.gitattributes`). Synthetic Tier-0 (or owner-cleared, de-attributed real). Drives the `golden_fixtures` characterization gate with **no video/GPU/DB**. See `docs/GOLDEN_REGRESSION_FIXTURES.md`; mint via `python -m backend.eval.golden_fixtures --freeze-synthetic` / `--freeze-real` (refusal-gated). |
38 | Pipeline run outputs (reels, trajectories, DBs, frames) | `output/` | **Gitignored.** Per-video isolation is the tracked **`PER_VIDEO_WORKSPACE.md`** project (P0 helper shipped #264; supersedes the archived `PER_VIDEO_OUTPUT_LAYOUT`). |
39
40 **Canonical files that stay at the repo ROOT** (referenced by CI / packaging / docs ? do not move):
41 `pyproject.toml` (PEP 621 packaging, hatchling; `indexer` entry point) · `run_all_tests.py` (CI: `ci.yml`) ·
42 `run_regression.py` (the no-regression gate CLI, cited in `EXPERIMENT_HARNESS.md`) · `run_eval.py`
43 (grandfathered ? imported by `tests/test_eval_harness.py`). The old diagnostic `setup.py` moved to
44 `scripts/doctor.py` (it shadowed the build system); invoke via `python scripts/doctor.py` or `indexer --setup`.
45 New standalone run-drivers go in **`scripts/`**, not the root.
46
47 ---
48
49 ## 1. PIPELINE
50
51 ### 1A. Runtime: video file -> highlight reel
52 ```
53 typed config load (built-in defaults -> config.json overrides; absent/parse-fail -> all-defaults, never fatal)
54 @backend/config/loader.py::load_config -> @backend/config/models.py::AppConfig
55 -> video_id = MD5(whole file) @backend/utils/hashing.py::compute_video_id (single shared helper, 64KB chunks; imported by main + all run_*.py ? no more per-entrypoint copies)
56 -> validate @backend/pipeline/validators/base.py::registry.run_all_with_context(video_path, global_config)
57 FormatValidator -> ctx['ffprobe_meta'] -> ResolutionFPSValidator -> PTSContinuity -> SceneCut -> Blur -> Relevance (? ACTUAL registration/exec order ? see @backend/pipeline/validators/__init__.py)

[validation FAIL -> persist status="failed", return early; report STILL emitted in finally]
-> db.add_video(video_id, filepath, status, [r.dict()]) @backend/infrastructure/database.py
-> seg_name = req.segmenter or global_config.indexing.default_segmenter (fallback 'motion')
-> seg = segmenter_registry.get(seg_name)(db, global_config) ? @backend/pipeline/segmenters/base.py::segmenter_registry (400 + available list if missing)
-> segments = seg.process_video(video_id, video_path, player_pool?, max_frames?)
[STOP POINT: segments-only ? predictions now in DB; eval/regression operate here without stitching]
-> finally: emit_indexer_report(...) @backend/reporting/__init__.py (? ALWAYS runs ? even on validation failure / exception; NEVER raises; grafts response["reports"])
-> select segment ids -> [(start,end)] (+ stitching padding)
-> VideoCompiler.compile_highlights(filepath, time_pairs, out_name) @backend/pipeline/stitcher/compiler.py
per-clip ffmpeg re-encode (libx264/superfast/crf22) -> concat -c copy -> output/<name>.mp4
```
Hybrid segmenter internal 4-phase (yolo_hybrid / tracknet_hybrid / wasb_hybrid ? same shape):
```
P1 chunk video (yolo only; tracknet/wasb = whole-video single pass, chunk_index=0)
-> P2 candidate "action windows" from motion/trajectory [STOP: candidates persisted via db.add_candidate_segment]
-> P3 pad clip (ai_handoff_padding) -> ThreadPoolExecutor -> provider.analyze_video(clip, prompt) ? provider_registry
-> P4 db.add_play_segment + db.link_candidate_to_segment
```
Pure-AI path `gemini`: no P2; chunk-or-single -> provider -> heavy validate (clamp/dedup) -> add_play_segment.
Legacy `motion`: pixel-diff -> segments + **fabricated** metadata/events (? not ground truth).

### 1B. WASB shuttle substrate (P2 of wasb_hybrid)
```

#### 15. assistant (2026-06-21T13:56:45.170Z)

khelsutra docs are all under `docs/` (flat, no NEXT_STEPS). Let me find the khelsutra cross-cutting register equivalent and confirm what tracks active docs they carry. Let me also check for a DOC_STATUS or similar in khelsutra and read the active tracked docs' status headers.

#### 16. user (2026-06-21T13:56:46.342Z)

C:\Users\avidu\Projects\khelsutra\README.md
C:\Users\avidu\Projects\khelsutra\docs\REQUEST_ACCESS_FORM.md
C:\Users\avidu\Projects\khelsutra\site\README.md
C:\Users\avidu\Projects\khelsutra\web\node_modules\@babel\parser\CHANGELOG.md
C:\Users\avidu\Projects\khelsutra\web\node_modules\@babel\compat-data\README.md
C:\Users\avidu\Projects\khelsutra\web\node_modules\@babel\helper-globals\README.md
C:\Users\avidu\Projects\khelsutra\web\node_modules\@babel\helper-module-imports\README.md
C:\Users\avidu\Projects\khelsutra\web\node_modules\@babel\helper-plugin-utils\README.md
C:\Users\avidu\Projects\khelsutra\web\node_modules\@babel\helper-string-parser\README.md
C:\Users\avidu\Projects\khelsutra\web\node_modules\@babel\helper-validator-identifier\README.md
C:\Users\avidu\Projects\khelsutra\web\node_modules\@babel\helper-validator-option\README.md
C:\Users\avidu\Projects\khelsutra\web\node_modules\@types\babel__generator\README.md
C:\Users\avidu\Projects\khelsutra\web\node_modules\@types\babel__template\README.md
C:\Users\avidu\Projects\khelsutra\web\node_modules\@types\babel__traverse\README.md
C:\Users\avidu\Projects\khelsutra\web\node_modules\@types\estree\README.md
C:\Users\avidu\Projects\khelsutra\web\node_modules\js-tokens\CHANGELOG.md
C:\Users\avidu\Projects\khelsutra\web\node_modules\@babel\template\README.md
C:\Users\avidu\Projects\khelsutra\web\node_modules\@babel\traverse\README.md
C:\Users\avidu\Projects\khelsutra\web\node_modules\@babel\types\README.md
C:\Users\avidu\Projects\khelsutra\web\node_modules\@babel\code-frame\README.md

```
C:\Users\avidu\Projects\khelsutra\web\node_modules\@babel\helper-compilation-targets\README.md
C:\Users\avidu\Projects\khelsutra\web\node_modules\@babel\helpers\README.md
C:\Users\avidu\Projects\khelsutra\web\node_modules\@babel\core\README.md
C:\Users\avidu\Projects\khelsutra\web\node_modules\@babel\generator\README.md
C:\Users\avidu\Projects\khelsutra\web\node_modules\@babel\helper-module-transforms\README.md
C:\Users\avidu\Projects\khelsutra\web\node_modules\@types\babel__core\README.md
C:\Users\avidu\Projects\khelsutra\web\node_modules\@types\deep-eql\README.md
C:\Users\avidu\Projects\khelsutra\web\node_modules\@rollup\rollup-win32-x64-gnu\README.md
C:\Users\avidu\Projects\khelsutra\web\node_modules\@rollup\rollup-win32-x64-msvc\README.md
C:\Users\avidu\Projects\khelsutra\web\node_modules\ms\license.md
C:\Users\avidu\Projects\khelsutra\web\node_modules\stackback\README.md
C:\Users\avidu\Projects\khelsutra\web\node_modules\@babel\parser\README.md
C:\Users\avidu\Projects\khelsutra\web\node_modules\ms\readme.md
C:\Users\avidu\Projects\khelsutra\web\node_modules\assertion-error\README.md
C:\Users\avidu\Projects\khelsutra\web\node_modules\check-error\README.md
C:\Users\avidu\Projects\khelsutra\web\node_modules\convert-source-map\README.md
C:\Users\avidu\Projects\khelsutra\web\node_modules\deep-eql\README.md
C:\Users\avidu\Projects\khelsutra\web\node_modules\lru-cache\README.md
C:\Users\avidu\Projects\khelsutra\web\node_modules\pathval\README.md
C:\Users\avidu\Projects\khelsutra\web\node_modules\tinyspy\README.md
C:\Users\avidu\Projects\khelsutra\web\node_modules\picocolors\README.md
C:\Users\avidu\Projects\khelsutra\web\node_modules\siginfo\README.md
C:\Users\avidu\Projects\khelsutra\web\node_modules\escalade\readme.md
C:\Users\avidu\Projects\khelsutra\web\node_modules\js-tokens\README.md
C:\Users\avidu\Projects\khelsutra\web\node_modules\lightningcss-win32-x64-msvc\README.md
C:\Users\avidu\Projects\khelsutra\web\node_modules\yallist\README.md
C:\Users\avidu\Projects\khelsutra\web\node_modules\strip-literal\README.md
C:\Users\avidu\Projects\khelsutra\web\node_modules\jsesc\README.md
C:\Users\avidu\Projects\khelsutra\web\node_modules\fdir\README.md
C:\Users\avidu\Projects\khelsutra\web\node_modules\@types\chai\README.md
C:\Users\avidu\Projects\khelsutra\web\node_modules\std-env\README.md
C:\Users\avidu\Projects\khelsutra\web\node_modules\@babel\plugin-transform-react-jsx-
source\README.md
C:\Users\avidu\Projects\khelsutra\web\node_modules\cac\README.md
C:\Users\avidu\Projects\khelsutra\web\node_modules\graceful-fs\README.md
C:\Users\avidu\Projects\khelsutra\web\node_modules\tinyglobby\README.md
C:\Users\avidu\Projects\khelsutra\web\node_modules\why-is-node-running\README.md
C:\Users\avidu\Projects\khelsutra\web\node_modules\semver\README.md
C:\Users\avidu\Projects\khelsutra\web\node_modules\update-browserslist-db\README.md
C:\Users\avidu\Projects\khelsutra\web\node_modules\@jridgewell\resolve-uri\README.md
C:\Users\avidu\Projects\khelsutra\web\node_modules\tinyrainbow\README.md
C:\Users\avidu\Projects\khelsutra\web\node_modules\@vitest\spy\README.md
C:\Users\avidu\Projects\khelsutra\web\node_modules\gensync\README.md
C:\Users\avidu\Projects\khelsutra\web\node_modules\tinybench\README.md
C:\Users\avidu\Projects\khelsutra\web\node_modules\tinyexec\README.md
C:\Users\avidu\Projects\khelsutra\web\node_modules\debug\README.md
C:\Users\avidu\Projects\khelsutra\web\node_modules\esbuild\LICENSE.md
C:\Users\avidu\Projects\khelsutra\web\node_modules\@types\react-dom\README.md
C:\Users\avidu\Projects\khelsutra\web\node_modules\estree-walker\README.md
C:\Users\avidu\Projects\khelsutra\web\node_modules\@tailwindcss\vite\README.md
C:\Users\avidu\Projects\khelsutra\web\node_modules\@types\react\README.md
C:\Users\avidu\Projects\khelsutra\web\node_modules\browserslist\README.md
C:\Users\avidu\Projects\khelsutra\web\node_modules\es-module-lexer\README.md
C:\Users\avidu\Projects\khelsutra\web\node_modules\esbuild\README.md
C:\Users\avidu\Projects\khelsutra\web\node_modules\node-releases\README.md
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C:\Users\avidu\Projects\khelsutra\web\node_modules\expect-type\README.md
C:\Users\avidu\Projects\khelsutra\web\node_modules\pathe\README.md
C:\Users\avidu\Projects\khelsutra\web\node_modules\strip-literal\node_modules\js-
tokens\README.md
C:\Users\avidu\Projects\khelsutra\web\node_modules\@tailwindcss\node\README.md
C:\Users\avidu\Projects\khelsutra\web\node_modules\@vitest\snapshot\README.md
C:\Users\avidu\Projects\khelsutra\web\node_modules\expect-type\SECURITY.md
C:\Users\avidu\Projects\khelsutra\web\node_modules\@vitest\runner\README.md
```

```
C:\Users\avidu\Projects\khelsutra\web\node_modules\picomatch\README.md
C:\Users\avidu\Projects\khelsutra\web\node_modules\baseline-browser-mapping\README.md
C:\Users\avidu\Projects\khelsutra\web\node_modules\csstype\README.md
C:\Users\avidu\Projects\khelsutra\web\node_modules\react-refresh\README.md
C:\Users\avidu\Projects\khelsutra\web\node_modules\tinypool\README.md
C:\Users\avidu\Projects\khelsutra\web\node_modules\@babel\plugin-transform-react-jsx-
self\README.md
C:\Users\avidu\Projects\khelsutra\web\node_modules\@vitejs\plugin-react\README.md
C:\Users\avidu\Projects\khelsutra\web\node_modules\electron-to-chromium\README.md
C:\Users\avidu\Projects\khelsutra\web\node_modules\json5\LICENSE.md
C:\Users\avidu\Projects\khelsutra\web\node_modules\@tailwindcss\oxide-win32-x64-msvc\README.md
C:\Users\avidu\Projects\khelsutra\web\node_modules\json5\README.md
C:\Users\avidu\Projects\khelsutra\web\node_modules\vite-node\README.md
C:\Users\avidu\Projects\khelsutra\web\node_modules\magic-string\README.md
C:\Users\avidu\Projects\khelsutra\web\node_modules\scheduler\README.md
C:\Users\avidu\Projects\khelsutra\web\node_modules\lightningcss\README.md
C:\Users\avidu\Projects\khelsutra\web\node_modules\@jridgewell\gen-mapping\README.md
C:\Users\avidu\Projects\khelsutra\web\node_modules\@jridgewell\sourcemap-codec\README.md
(Results are truncated. Consider using a more specific path or pattern.)
```

**17. user (2026-06-21T13:56:47.535Z)**

```
C:\Users\avidu\Projects\khelsutra\docs\REQUEST_ACCESS_FORM.md:3:> **Status:** `IN PROGRESS` ·
**Owner:** Avi · **Created:** 2026-06-20 · **Last updated:** 2026-06-20
C:\Users\avidu\Projects\khelsutra\docs\FRIENDLY_AND_MULTILINGUAL.md:3:> **Status:** `DRAFT` ?
owner-gated (implementation starts only on explicit approval) · **Owner:** Avi Dullu ·
**Created:** 2026-06-21 · **Last updated:** 2026-06-21
C:\Users\avidu\Projects\khelsutra\docs\LOCAL_APPLIANCE_ARCHITECTURE.md:3:> **Status:** `DRAFT` ?
owner-gated (auth choice, retention, cost ceiling, GPU host, P5 bridge) · **Owner:** Avi ·
**Created:** 2026-06-20 · **Last updated:** 2026-06-20
C:\Users\avidu\Projects\khelsutra\docs\DRY_RUN_CUJ.md:3:> **Status:** `DRAFT` ? owner-gated ·
**Owner:** Avi Dullu · **Created:** 2026-06-21 · **Goal for this session.**
C:\Users\avidu\Projects\khelsutra\docs\PER_RALLY_SELECTION.md:3:> **Status:** `IN PROGRESS` ?
**MVP / M0 (P1?P5) COMPLETE 2026-06-20** (pick ? query ? select ? compile ? download, on
existing engine endpoints; verified e2e against a stub engine, GCP-GPU validation next). Design
**APPROVED by owner 2026-06-20**. Downstream milestones M1?M5 (multi-video auth, per-rally
preview endpoint, corpus loop, localized NL) remain owner-gated. · **Owner:** Avi · **Created:**
2026-06-20 · **Last updated:** 2026-06-20
C:\Users\avidu\Projects\khelsutra\docs\ANNOTATE_AND_TRAIN.md:3:> **Status:** `DRAFT` ? owner-
gated on scope · **Owner:** Avi · **Created:** 2026-06-20 · **Last updated:** 2026-06-20
C:\Users\avidu\Projects\khelsutra\docs\PORTABILITY.md:3:> **Status:** design + proven
(2026-06-20). **Decision:** keep the site/form host-portable so
```

**18. assistant (2026-06-21T13:56:58.348Z)**

The khelsutra docs README doesn't index `ANNOTATE_AND_TRAIN.md` or `PORTABILITY.md` is indexed but `ANNOTATE_AND_TRAIN.md` is not in the README. Let me verify the khelsutra git state (it's on master with untracked `.claude/` and `site/access.db` only, so master is clean) and check the engine git divergence to understand the multi-branch pull issue. Let me also confirm whether the engine's serving milestone commits are actually merged to master.

**19. user (2026-06-21T13:57:03.140Z)**

```
=== ENGINE: local branches ===
+ bhi-execpolicy                            5cb7bbf (C:/Users/avidu/Projects/badminton-
highlight-indexer/.claude/worktrees/bhi-execpolicy) [origin/bhi-execpolicy: gone] feat(jobs):
worker self-scheduling drain loop honoring WAIT_AND_RESUME (EXECUTION_POLICY P3a)
  chore/consolidate-fusion-schema-predicate e268972 [origin/chore/consolidate-fusion-schema-
predicate: gone] refactor(eval): single source of truth for the _is_fusion_schema predicate
  docs/adr-012-gpu-cloudrun                 118d9b2 [origin/docs/adr-012-gpu-cloudrun: gone]
docs(adr): ADR-012 ? GPU-native compute; TPU deferred; Cloud Run+GPU is the serving model
  docs/archive-decoupled-compute            e6110ce [origin/master: behind 21] feat(api): GET
/api/health + return download_url from /api/compile (M1/P6) (#268)
  docs/archive-decoupled-v2                 fb51cdc [origin/docs/archive-decoupled-v2: gone]
```

```
docs: archive DECOUPLED_COMPUTE ? DELIVERED (M0-M2 + M3.5 on GCP)
  docs/compute-decoupled-serving            a60da27 [origin/docs/compute-decoupled-serving: gone] docs: spin up COMPUTE_DECOUPLED_SERVING subproject + ADR-014
  docs/court-floor-rally-end                430de76 [origin/docs/court-floor-rally-end: gone] docs(rally): capture the shuttle-floor-contact rally-END signal (owner note)
  docs/decouple-scope-defer                 4b2c202 [origin/docs/decouple-scope-defer: gone] docs(decoupled): narrow DECOUPLED_COMPUTE scope (M0?M2 + M3.5); defer M3/M4; DO=train, GCP=serve (ADR-011)
  docs/deploy-report-polish                 7cc2300 [origin/docs/deploy-report-polish: gone] docs(deploy): report polish ? highlight reel smoke-test result + VM-deleted final state
  docs/desktop-app-plan                     b9ff235 [origin/docs/desktop-app-plan: gone] docs(desktop): scope a cross-platform desktop app + the de-WSL native-compute port
  docs/doc-status-register                  b1c0f37 [origin/docs/doc-status-register: gone] docs: add DOC_STATUS register + archival due-diligence convention
  docs/drop-digitalocean-gcp-only           5225bc6 [origin/docs/drop-digitalocean-gcp-only: gone] docs: drop DigitalOcean ? GCP is the sole compute stack (ADR-013)
  docs/findings-1-2-pervideo-nextsteps      190a2c8 [origin/docs/findings-1-2-pervideo-nextsteps: gone] docs: resolve DOC_STATUS findings #1 (per-video supersede) + #2 (NEXT_STEPS refresh)
  docs/gcp-deploy-report                    76ae710 [origin/docs/gcp-deploy-report: gone] docs(deploy): first real GCP-L4 run report + gpu_setup conda-ToS/deps fix + Ops Agent runbook
  docs/gcp-pivot-runbook                    e2e9754 [origin/docs/gcp-pivot-runbook: gone] docs(gcp): add Spot-T4 fast path (likely skips the GPUS_ALL_REGIONS wait)
  docs/golden-regression-fixtures-plan      f5f5e52 [origin/docs/golden-regression-fixtures-plan: gone] feat(eval): R5 ? blob-fallback splits the continuously-active blob (default-OFF) (#203)
  docs/lock-m0-decisions                    21c93d1 [origin/docs/lock-m0-decisions: gone] docs(serving): lock the 5 M0 gating decisions (D7-D12) + add the data-flywheel harness (M11)
+ docs/lovo-resweep-n13                     fe34013 (C:/Users/avidu/Projects/badminton-highlight-indexer/.claude/worktrees/resweep) [origin/docs/lovo-resweep-n13: gone] docs(archive): make lovo_resweep.py lint-clean (def/rename/noqa E402)
  docs/m0-approved-d13-d15                  93d2188 [origin/docs/m0-approved-d13-d15: gone] docs(serving): M0 APPROVED + lock D13-D15 (switchable profiles, Drive Northstar, consent-not-surprise)
  docs/m2-finalized-report                  8d8d230 [origin/docs/m2-finalized-report: gone] docs(deploy): M2 FINALIZED ? Gen-0 weights/0.1.0-l4 pushed; both VMs deleted ($0)
=== ENGINE: master vs origin/master ===
e68b583 fix(api): resolve a bucket-ingested video's source to a local file before stitching (#289)
43eccc1 feat(serving): M8 ? per-job cost ledger + versioned pricebook + /cost API (default-OFF) (#288)
8cdf65c feat(serving): M6 ? Cloud Run GPU dispatch shim + worker image (default-OFF, mocked) (#287)
=== ENGINE: is M4 serving on master? ===
43eccc1 feat(serving): M8 ? per-job cost ledger + versioned pricebook + /cost API (default-OFF) (#288)
8cdf65c feat(serving): M6 ? Cloud Run GPU dispatch shim + worker image (default-OFF, mocked) (#287)
b5af66e feat(serving): M5 ? truly-local profile per-profile startup checks + L4 profile-parity test (default-OFF) (#286)
fefa412 docs(serving): resolve §8 design picks (D16/D17) + reconcile per-video P3 vs M3 (#285)
cee4baa feat(serving): M4 ? DeviceContext + native WASB CPU-fallback (device="auto", cuda stays strict) (#283)
2afd54f feat(serving): M3 ? configurable output/scratch base (de-hardcode output/, default-OFF) (#282)
e111add feat(serving): M2 ? URI-aware input (input_backend/input_root, local=identity) (#280)
ce22791 feat(serving): M1 ? deployment.profile + compute_target config (pure, additive, default-OFF) (#279)
f518e97 docs(serving): M0 APPROVED + lock D13-D15 (switchable profiles, Drive Northstar, consent-not-surprise) (#278)
49c082a docs(serving): lock the 5 M0 gating decisions (D7-D12) + add the data-flywheel harness (M11) (#277)
=== ENGINE: COMPUTE_DECOUPLED_SERVING dir present? ===
README.md
TESTING_STRATEGY.md
```

architecture.svg
runlogs

### 20. assistant (2026-06-21T13:57:12.223Z)

This is a key finding: origin/master is significantly AHEAD of what DOC_STATUS and NEXT_STEPS describe. DOC_STATUS (last reviewed 2026-06-21) says "M1?M4 SHIPPED" and "M5+ batch authorized", but origin/master already has **M5 (#286), M6 (#287), M8 (#288), and a bucket-ingest fix (#289)** merged. The serving track has progressed well past what the registers claim. Let me confirm the full set of merged serving milestones and check the COMPUTE_DECOUPLED_SERVING tracker §7 to see if it was updated.

### 21. user (2026-06-21T13:57:13.325Z)

```
=== All serving + per-video commits on origin/master (recent) ===
e68b583 fix(api): resolve a bucket-ingested video's source to a local file before stitching
(#289)
43eccc1 feat(serving): M8 ? per-job cost ledger + versioned pricebook + /cost API (default-OFF)
(#288)
8cdf65c feat(serving): M6 ? Cloud Run GPU dispatch shim + worker image (default-OFF, mocked)
(#287)
b5af66e feat(serving): M5 ? truly-local profile per-profile startup checks + L4 profile-parity
test (default-OFF) (#286)
fefa412 docs(serving): resolve §8 design picks (D16/D17) + reconcile per-video P3 vs M3 (#285)
06ff4b7 test(api): lock gs://|gcs:// /api/process -> queryable API-DB video_id (B-P1 guard)
(#284)
cee4baa feat(serving): M4 ? DeviceContext + native WASB CPU-fallback (device="auto", cuda stays
strict) (#283)
2afd54f feat(serving): M3 ? configurable output/scratch base (de-hardcode output/, default-OFF)
(#282)
41d45fd fix(server): drain async jobs under `python -m backend.main` (double-module hang) (#281)
e111add feat(serving): M2 ? URI-aware input (input_backend/input_root, local=identity) (#280)
ce22791 feat(serving): M1 ? deployment.profile + compute_target config (pure, additive, default-
OFF) (#279)
f518e97 docs(serving): M0 APPROVED + lock D13-D15 (switchable profiles, Drive Northstar,
consent-not-surprise) (#278)
49c082a docs(serving): lock the 5 M0 gating decisions (D7-D12) + add the data-flywheel harness
(M11) (#277)
f592281 docs: point I18N_PLAN + UI_PROPOSAL (+ DOC_STATUS) at the khelsutra UX/l10n project
(#275)
f6f15ca docs: isolate field-ops/PMF + Roora into archives/; carve a no-ML docs/data_pipeline/
(#276)
03fa951 docs: resolve DOC_STATUS findings #1 (per-video supersede) + #2 (NEXT_STEPS refresh)
(#274)
aa5e9db docs: add DOC_STATUS register + archival due-diligence convention (#273)
bd42250 docs: spin up COMPUTE_DECOUPLED_SERVING subproject + ADR-014 (#272)
4957b72 fix(worker): configure logging + explain a 0-segment infer (the cold-start blind spot)
(#271)
4134721 docs: archive DECOUPLED_COMPUTE ? DELIVERED (M0-M2 + M3.5 on GCP) (#270)
c8e60ee fix(gen0): provenance-stamp the manifest git_sha off a git-archive tarball (#269)
e6110ce feat(api): GET /api/health + return download_url from /api/compile (M1/P6) (#268)
a555887 docs(deploy): M2 FINALIZED ? Gen-0 weights/0.1.0-l4 pushed; both VMs deleted ($0) (#267)
b05401b docs: drop DigitalOcean ? GCP is the sole compute stack (ADR-013) (#266)
4d546ec fix(gcp): teardown defaults to DELETE (a stopped VM still bills its disk) (#265)
dd30e9b feat(workspace): per-video workspace convention ? design doc + Phase 0 helper (#264)
dc9111b docs(adr): ADR-012 ? GPU-native compute; TPU deferred; Cloud Run+GPU is the serving
model (#263)
d8c5aa0 docs(deploy): report polish ? highlight reel smoke-test result + VM-deleted final state
(#262)
0d73d96 docs(deploy): first real GCP-L4 run report + gpu_setup conda-ToS/deps fix + Ops Agent
runbook (#261)
380a4d0 docs(decoupled): narrow DECOUPLED_COMPUTE scope (M0?M2 + M3.5); defer M3/M4; DO=train,
GCP=serve (ADR-011) (#260)
36ed29f feat(wasb): native runner defaults to streaming + correct M3.5 parity gate to a
tolerance (#259)
```

```
38c2d0f feat(storage): gdrive:// reads a locally MOUNTED Google Drive (no API) (#258)
55ae408 docs(decoupled): GCP pivot runbook + DO?GCP ADR-010 (deploy/gcp + handoff) (#257)
dc53db5 feat(detectors): Linux-native WASB runner (M3.5) + worker real-trajectory extraction
(#256)
5c77f48 docs(decoupled): HANDOFF.md ? session checkpoint / resume-here (#255)
d5407eb feat(worker): Gen-0 train-from-bucket (worker train) on a CPU stub-trajectory mock
(#254)
814d813 feat(deploy): GPU machine provisioning ? NVMe scratch + native WASB setup (#253)
f06c6a6 fix: cross-platform canonicalization (Linux/macOS, not just Windows) (#252)
6e5ac9e feat(worker): python -m backend.worker dispatcher ? bucket-driven train/infer (#251)
d292854 docs: SETUP_NEW_MACHINE ? canonical zero?train/infer runbook (#250)

=== current checked-out HEAD vs origin/master (am I behind?) ===
1       0
```

### 22. assistant (2026-06-21T13:57:18.195Z)

```
Confirmed. origin/master is well ahead of the registers. The serving track has shipped **M1?M6 +
M8** plus the bucket-ingest B-P1/B-P2 fixes (#284, #289). DOC_STATUS (last review 2026-06-21)
only credits M1?M4. Let me check the COMPUTE_DECOUPLED_SERVING §7 tracker on origin/master to
see if it tracks the truth, and verify the I18N/UI cross-repo pointer rows. Let me read the
serving tracker from origin/master.
```

### 23. user (2026-06-21T13:57:20.676Z)

```
5:> **Tracking anchors:** §7 progress tracker is the source of truth; pointer in memory
`current-state` + `docs/NEXT_STEPS.md` once work starts.
6-> **Relation to existing docs:** the productionization successor to the archived
`DECOUPLED_COMPUTE` (M3/M4); realizes the `ComputeBackend`/`StorageBackend` registries from
[PLATFORM_ARCHITECTURE.md](../PLATFORM_ARCHITECTURE.md) §3; the **truly-local** profile is the
L0 tier of [VIDEO_LOCALITY_MODEL.md](../VIDEO_LOCALITY_MODEL.md) and the `LocalDesktop` backend
of [DESKTOP_APP_PLAN.md](../DESKTOP_APP_PLAN.md).
7-> **Naming:** this was the "Cloud Run + GPU serving subproject" (ADR-012). Renamed **compute-
decoupled serving** because **truly-local is a co-equal, first-class profile**, not an
afterthought.
8-> **Honesty note:** claims tagged `[verified]` (checked against code by the design workflow
`wf_56274ff0`), `[design]` (proposed), `[researched]`.
9-
10----
11-
12-> **? Northstar (D14):** a **mobile, app-like** flow ? point at a **Google Drive** video ?
**fully cloud-native** run ? the stitched, **ready-to-share** reel lands **back in Drive, next
to the source**. Two design promises around it: a user can **switch local?cloud anytime** (D13,
a headline feature), and execution is **never a surprise** ? auto-detect *suggests*, the user
*confirms* (D15).
13-
14-## 0. TL;DR
15-The rally engine is a **pure function of `(input bytes + config) ? outputs`** in three
deterministic stages ? **DETECT ? WINDOW ? STITCH** ? that **no deployment profile may branch
on**. A **`ComputeBackend` (deployment profile)** decides *where* the core runs (a local process
vs a Cloud Run GPU job); a **`StorageBackend`** decides *where bytes come from* (local FS / USB
/ mounted Google Drive / bucket). **One startup config** picks both. We ship **truly-local**
(local GPU + in-process stitch, zero cloud) and **cloud-serving** (upload `<2GB` / gdrive /
bucket ? a Cloud Run GPU job runs detect **and** stitch, so stitch's compute is metered), plus
**cpu-mock** for CI. The single most important caveat: detection is **byte-identical only on the
same GPU** (cross-GPU is sub-pixel-different, per the 2026-06-20 deploy report) ? so the
**parity guarantee is enforced at the WINDOW + STITCH layer**, not detector-CSV bytes.
16-
17-## 1. Problem & goal
18-- **Why now:** DECOUPLED_COMPUTE proved compute *portability* (M0?M2 + M3.5 on a GCP L4) and
is archived; its deferred **M3 (serving endpoint + per-video billing) + M4 (scale-to-zero + cost
guard)** become this project. The owner needs **both** a hosted service *and* a truly-local
mode, with the **core (detect+stitch) unaffected** by which one runs.
19-- **The two user-facing outcomes** (owner, 2026-06-21):
20-  - **(a) Hosted service** ? operate on small **local uploads (`<2GB`)** *and* load from
```

**gdrive / a bucket**, spinning up **Cloud Run for both rally segmentation AND stitching** (stitching is compute-intensive ? run it where its cost is accounted for).
21-  - **(b) Truly-local** ? on a local machine with videos on **local drives/USB + a local GPU**, run inference **and** stitching **completely locally**.
22-- **"Good" looks like:** the same `video ? rallies ? reel` core gives identical windows + stitched ranges whether it runs on a laptop or Cloud Run; switching is a **config/startup choice**, never a code branch in the core.
23-
24-## 2. Decisions locked
25-
26-| # | Decision | Source | Implication |
27-|---|----------|--------|-------------|
28-| D1 | The core is a **pure 3-stage function** (DETECT/WINDOW/STITCH); **no profile branch inside it**. | design `[verified]` | All profile difference lives in config + the two registries. |
29-| D2 | **Two registries:** `ComputeBackend` (where) selected by `deployment.profile` + `compute_target`; `StorageBackend` (what-bytes) by `input_backend`/`storage.backend`. | ADR-014 | Realizes PLATFORM_ARCHITECTURE's `local\|hosted\|byo_worker`. |
30-| D3 | **Profiles shipped:** `truly-local`, `cloud-serving`, `cpu-mock` (CI); `byo_worker` reserved/deferred. | ADR-014 | Enum/seam reserved so BYO isn't precluded. |
31-| D4 | In **cloud-serving, STITCH runs on Cloud Run** (co-located with detect) so CPU/wall + egress are metered into the per-job cost ledger. | owner (a) | Stitch-on-laptop would hide real cost. |
32-| D5 | **Parity enforced at WINDOW+STITCH** (byte-exact for a fixed trajectory); **detect** is byte-exact only same-GPU. | deploy report 2026-06-20 | Cross-GPU asserted at the rally/window level, never CSV bytes. |
33-| D6 | **Default-OFF, smallest-safe-first, one PR per milestone**; any core-caller change (e.g. configurable output base) ships behind a flag + a Launch Report. | [[launch-report-convention]] | Zero-regression discipline. |
34-| D7 | **Cloud Run cold-start: SCALE-TO-ZERO** ? accept the ~1?3 min cold-start (it's an async job: upload?poll?reel, amortized over the multi-minute job). **No warm pool** (that ~$500/mo idle would break per-video billing). | owner 2026-06-21 | M6/M7. Revisit a warm lane only on a UX complaint. |
35-| D8 | **Pricebook = a versioned DB table**; cost computed in **USD** (matches GCP billing = one source of truth), **displayed in INR** at a configurable FX rate (Bangalore market); re-versioned when GCP prices change. | owner 2026-06-21 | M8. |
36-| D9 | **Edge auth = Cloudflare Access** (reuse the site's existing CF). Lock the Cloud Run **ingress to the CF proxy** + **verify the `Cf-Access` JWT signature** (so a direct hit to the Cloud Run URL can't spoof the identity header). | owner 2026-06-21 | M9. One identity system. |
37-| D10 | **Free-tier "data-for-reels" trade:** free reels in exchange for **training rights** (uploaded footage + derived trajectories/labels); **paid tier = no-train** (privacy is the paid feature). **Carve-outs:** train ONLY on attested **ADULT** footage (minor footage ? reel yes, training pool **NO** ? DPDP/GDPR need verifiable parental consent); train on **DERIVED** data + **auto-delete raw**; **ToS counsel-reviewed**; live training-on-uploads **gated to M9** (public signup). Truly-local needs no DPA. | owner 2026-06-21 | M9 + D12; ties [[paper-coauthor-aim]] / PERSONALIZATION flywheel. |
38-| D11 | **Quality floor: Gemini handoff ON** for cloud-serving until the owned model clears the #172 ship-gate (e.g. ?0.70 multi-court) ? then flip via a **Launch Report**; the owned model runs in **shadow/eval** meanwhile. (Gemini = serving/inference ONLY, never a training source ? CLAUDE.md guardrail.) | owner 2026-06-21 | M7 + D12. |
39-| D12 | **Data-flywheel harness is IN SCOPE (owner add, Q5):** capture the (consented, adult) uploaded video + the Gemini reel/rally output ? **reliably store** in the per-video workspace/bucket ? **run shadow evals** (owned-vs-Gemini, dual-eval-set) ? **promote good ones to the golden TRAINING set** (human-reviewed labels) so the owned model climbs toward the D11 floor (then Gemini flips off). | owner 2026-06-21 | new **M11**; ties D10 (consent) + `data_pipeline/` + the eval/experiment harness. |
40-| D13 | **Profile is user-switchable ANYTIME ? a headline FEATURE.** The same user runs **local today, cloud tomorrow, back to local** ? it's just a config/startup choice; the pure core gives **equivalent** output either way. **Advertise it** ("your machine *or* our cloud ? your call, switch anytime"). | owner 2026-06-21 | Honest caveat: equivalent at the **rally/window** level, not bit-identical across GPUs (D5); a single job runs in one profile, switching is *between* jobs. |
41-| D14 | **NORTHSTAR ? operate toward this:** a **mobile, app-like** flow ? point at a **Google Drive** video ? **fully cloud-native** run ? the stitched, **ready-to-share** reel lands **back in Drive, next to the source**. | owner 2026-06-21 | Implies a **`gdrive-api`

(OAuth) StorageBackend** (read source + write reel back) ? distinct from the local **mount**
backend; **resolves §8 #7**, new **M12**. The mobile UI is the khelsutra track; the engine must
support it (async jobs + status + Drive round-trip / signed URL). Net-new scope (OAuth + Drive
write); not necessarily v1, but the design must not preclude it (the `StorageBackend` ABC
accommodates it). |
42-| D15 | **Auto-detect SUGGESTS, the user CONFIRMS ? execution is NEVER a surprise.** A GPU
owner may still choose cloud; **cloud (= spend) is never entered silently**; the active profile
is always explicit + surfaced. | owner 2026-06-21 | Amends §4.3 (auto-detect = a proposed
default, not an auto-decision). |
43-| D16 | **M6/M7 implementation SHAPE + the cheap §8 picks (owner 2026-06-21):** (a) **Cloud
Run JOBS** (not Services-with-GPU) for the detect+stitch unit, **bucket-poll for completion**
(reuses `JobWaiting`/`claim_due_job` verbatim, fits async upload?poll?reel + scale-to-zero D7);
(b) **stitch ENCODE = libx264** (protects the D5 byte-determinism/parity guarantee + the L4
parity test) ? **NVDEC for DECODE only** (a speed lever that never touches output bytes);
**NVENC encode is DEFERRED behind a config knob**, revisit only if M7 cost data justifies
relaxing parity (§8 Q8); (c) **input cap = hard-reject `<2GB`** for v1 (a config cap so raising
to 4GB later is one line; matches the shipped M2 chunked-upload threshold ? §8 Q9). | owner
2026-06-21 | M6/M7. Resolves §8 Q2 (co-located, = D4), Q8, Q9, Q10 (dir name confirmed). |
44-| D17 | **Real cloud verification BUDGET = $100 / ?10k cap (owner 2026-06-21):** the owner
authorized actual GCP spend for due-diligence + production-grade testing/verification of M5?M7
(real L4 run, the M5 runlog, the M7 DEPLOY_REPORT), **provided the cap is respected** ? confirm
exact cmd + hourly cost before each spend, prefer the smallest viable run + cheapest-adequate
GPU, and **DELETE every resource after** ([[gcp-vm-always-delete]]). Removes the M6/M7 "spend"
gate within the envelope; the calibrated pricebook (M8) + counsel ToS (M9) + OAuth app (M12)
stay owner/counsel-side. | owner 2026-06-21 | Unblocks the real M5/M7 runs (not just the code)
within the cap. |
45-
--
110:## 7. Deliverables & progress tracker  ? **source of truth**
111-Legend: ? Todo · ? In progress · ? Done · ? Gated. **One small PR per row**, off
`origin/master`, no stacking. Default-OFF where it touches core callers.
112-
113-| ID | Deliverable | Depends | Gated? | Status | PR |
114-|----|-------------|---------|--------|--------|----|
115:| M0 | This design doc + project dir + `runlogs/` + ADR-014 | ? | owner sign-off | ? | **?
APPROVED 2026-06-21** |
116:| M1 | `deployment.profile` + `compute_target` config + preset bundles + precedence/auto-
detect; unit tests (no behaviour change, profile='' default) | M0 | safe | ? |
[#279](https://github.com/Khelsutra/badminton-highlight-indexer/pull/279) |
117:| M2 | `input_backend`/`input_root`; wire main CLI/API input through `StorageRef.parse_uri`
(local=identity); storage-fake tests | M1 | safe | ? |
[#280](https://github.com/Khelsutra/badminton-highlight-indexer/pull/280) |
118:| M3 | Configurable output/scratch base ? kill hardcoded `output/`
(`trajectory_hybrid:237,313`, `yolo_hybrid:407`, `gemini`); **DEFAULT stays `output/`**; golden
+ stub-E2E byte-identical | M1 | ? default-OFF + Launch Report on flip | ? |
[#282](https://github.com/Khelsutra/badminton-highlight-indexer/pull/282) |
119:| M4 | `DeviceContext` + WASB **CPU-fallback**; unified healthcheck/device wiring | M1 |
safe (cuda default unchanged) | ? | [#283](https://github.com/Khelsutra/badminton-highlight-
indexer/pull/283) |
120:| M5 | **truly-local profile** end-to-end (preset + startup checks); first committed
**runlog** (truly-local sample run) | M2,M3,M4 | owner real-GPU | ? |
[#286](https://github.com/Khelsutra/badminton-highlight-indexer/pull/286) (preset+startup-
checks+L4 parity ?; runlog ? owner GPU) |
121:| M6 | Cloud Run GPU **dispatch shim** (control plane ? job via storage ? re-claim) +
**containerized worker image** (ffmpeg+CUDA+WASB+fonts); mocked-dispatch tests | M5 | ? owner
(GCP spend) | ? | [#287](https://github.com/Khelsutra/badminton-highlight-indexer/pull/287)
(shim+image+mocked ?; real build/run = M7 ? owner) |
122:| M7 | **cloud-serving profile** e2e on L4: upload`<2GB` + gdrive/bucket ? detect+stitch on
Cloud Run ? reel via signed URL; **DEPLOY_REPORT** (posterity, sample runs + contact sheet) | M6
| ? owner (spend) | ? | ? |
123:| M8 | Per-job **cost ledger** (v6 migration: `owner_id, gpu_active_seconds, wall_seconds,
bytes_out, cost_usd, cost_breakdown_json`) + **pricebook** + aggregation + `/api/.../cost` | M7
| ? owner (billing) | ? | [#288](https://github.com/Khelsutra/badminton-highlight-
indexer/pull/288) (ledger+pricebook+/cost ? default-OFF; real prices + usage-emission ?
owner/follow-up) |

124:| M9 | **Edge auth** (Cf-Access extraction) + per-tenant scoping on `user_id`; DPA/consent gate wiring | M7 | ? owner (privacy/counsel) | ? | ? |
125:| M10 | `byo_worker` / zero-infra-BYO ? **design-only, DEFERRED** | M8,M9 | ? explicit owner approval | ? | ? |
126-| **M11** | **Data-flywheel harness (D12):** on a consented adult job, **capture** the upload + Gemini reel/rally output ? **reliably store** under the per-video workspace ? **shadow-eval** owned-vs-Gemini (dual-eval-set / `experiment.compare`) ? **promote** good ones to the golden TRAINING set (human-reviewed labels via the annotation flow). Closes the loop so the owned model climbs to the D11 floor. Reuses `data_pipeline/` + `backend/eval/` + the consent gate (M9). | M7,M9 | ? owner (consent/privacy) | ? | ? |
127-| **M12** | **`gdrive-api` (OAuth) StorageBackend (D14 Northstar):** cloud reads the source from + writes the stitched reel **back to the user's Google Drive next to the source** (per-user OAuth consent) ? the mobile/cloud-native Drive round-trip. Distinct from the local mount backend; slots into the `StorageBackend` ABC. | M7,M9 | ? owner (OAuth/consent) | ? | ? |
128-
129-**Testing strategy is a first-class deliverable** ? [`TESTING_STRATEGY.md`](TESTING_STRATEGY.md). **Run logs / sample runs for posterity** ? [`runlogs/`](runlogs/).
130-
131-## 8. Open questions ? owner *(blocking-gates vs nice-to-knows)*
132-**? The 5 gating questions are LOCKED (owner, 2026-06-21) ? see §2 D7?D12:**
133-1. ? **Cold-start ? D7** (scale-to-zero, accept cold-start).
134-3. ? **Pricebook ? D8** (versioned DB table; USD internal / INR display).
135-4. ? **Edge auth ? D9** (Cloudflare Access; verify Cf-Access JWT, lock ingress to proxy).
136-5. ? **DPA/consent ? D10** (free-tier data-for-reels trade; adults-only; derived-data; counsel-gated to M9).
137-6. ? **Quality floor ? D11** + the **data-flywheel harness D12 / M11** (Gemini-ON until the owned model clears the floor; capture?store?eval?goldenize meanwhile).

### 24. assistant (2026-06-21T13:57:28.190Z)

This is the crucial finding: the COMPUTE_DECOUPLED_SERVING §7 tracker (source of truth, on origin/master) is MORE current than DOC_STATUS. It shows M5/M6/M8 as ? in-progress (code shipped, real-cloud/owner residuals pending), while DOC_STATUS §3a row 33 only says "M1?M4 SHIPPED; M5+ batch authorized." The tracker is accurate; DOC_STATUS is stale on this row.

Now let me verify a couple more potential staleness flags: the khelsutra `PER_RALLY_SELECTION` status (DOC_STATUS calls UI_ALPHA "ACTIVE"), the `DRY_RUN_CUJ` reel-download claim vs the khelsutra commit `#57` saying "Path-B live e2e DONE", and whether ANNOTATE_AND_TRAIN is tracked anywhere. Let me check the local working tree state (engine is on a different branch than the snapshot) and the khelsutra DRY_RUN status.

### 25. user (2026-06-21T13:57:31.035Z)

=== KHELSUTRA recent master commits ===
7750afc docs(dry-run): Path-B live e2e DONE ? in-UI bucket?reel CUJ + screencast (DoD §6 box 3 ?) (#57)
4525c5d feat(web): B-P2 ? in-UI bucket ingest (point at a gs:// video ? process ? load) (#56)
d0461fd docs: correct stale "bucket ingest is net-new" claims (B-P1 already shipped in M2/#280) (#55)
2421557 docs(dry-run): record Path-A local result (en+hi reel downloaded) + the engine fix it surfaced (#54)
113587f docs: set the dry-run CUJ as a tracked goal (UI drives a bucket video -> reel download, multilingual) (#53)
1b69e1a feat(site): show a localized UI screenshot on each academy flyer (#52)
3a42847 feat(web): localized sample-query prompt (en/kn/hi) + i18n-test rule for query parsing (#51)
0021c4f test(web): guard the SPA stays self-contained / offline-renderable (#50)
631c3aa docs(friendly-multilingual): SPA fully trilingual ? mark B1/B2/B4/B5/B6 done + screencast milestone (#49)
ad0d7ed feat(web): ship the Hindi (hi) catalog ? tracker B6 (the extensibility proof) (#48)
4449981 feat(web): ship the Kannada (kn) catalog ? tracker B5 (#47)
a423b82 feat(web): self-host OFL Noto Kannada + Devanagari fonts ? tracker B4 (#46)

=== DRY_RUN_CUJ DoD / status lines ===

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docs/DRY_RUN_CUJ.md:1:# Dry-run CUJ ? UI drives the tool on a bucket video ? reel downloaded local (multilingual)
docs/DRY_RUN_CUJ.md:11:**? Back half (shipped ? PER_RALLY_SELECTION M0):** the SPA loads `?video=<id>` ? `POST /api/query` (rallies, default all-selected) ? deterministic query/select ? `POST /api/compile` ? `<video>` preview + **`<a download>` saves the MP4 locally**. Reel download-to-local **works today** (`GET /api/highlights/{video_id}/{filename}` streams it; `/api/compile` also returns `download_url`). Fully **en/kn/hi** (the switcher / `?lang=`).
docs/DRY_RUN_CUJ.md:14:- The SPA has **no ingest/process UI** ? it can't point at a video (bucket or local), can't call `/api/process`, can't poll `/api/jobs`. It only gets a `video_id` from `?video=` in the URL. **This SPA front-half is the *only* net-new work for Path B (= B-P2).**
docs/DRY_RUN_CUJ.md:15:- **HTTP bucket-ingest ALREADY EXISTS in the engine (shipped M2 / [#280]) `[verified file:line]`:** `POST /api/process` accepts a `gs://`/`gcs://` (also `s3://`/`local://`) `video_path` directly. Submit resolves the URI and **skips the local-existence check for non-local backends** (`backend/main.py:274,283` ? `storage.resolve_input_ref`; `gs://`?`gcs` normalized in `backend/storage/ref.py:60`); the job worker then fetches it to scratch (`_execute_processing` ? `storage.acquire_local_input`, `main.py:159`) and runs the **same** hash ? validate ? segment ? DB pipeline as a local file ? writing a queryable `video_id` into the API's SQLite `videos`/`play_segments`. Proven end-to-end by `tests/test_api_endpoints.py::test_process_nonlocal_uri_skips_exists_and_fetches` (gs:// ? 202, not 404 ? job `done` ? `db.get_video(video_id)` row with the **source URI preserved**), plus the hosted-mode-rejection / unknown-scheme-400 / CLI-variant tests.
docs/DRY_RUN_CUJ.md:21:**Path B ? Full "point at a bucket video IN the UI" (SPA-only ? smaller than first scoped).** The **HTTP bucket-ingest endpoint already exists** (§1) ? so Path B is **only the SPA front-half** (appliance S1/S2): an **ingest** input (enter a `gs://` URI) ? `POST /api/process` ? an **ingest + progress** screen polling `GET /api/jobs/{id}` ? on `done`, load `result.video_id` (which the existing back-half already renders). **No net-new engine endpoint** ? B-P1 is already done; the remaining work is **B-P2 (the SPA)**.
docs/DRY_RUN_CUJ.md:34:1. **Scope:** Path A (cheapest dry run now) or Path B (the in-UI bucket ingest ? **SPA-only**; the engine endpoint already accepts `gs://`, see §1)? *Recommend A first* (proves the multilingual workflow fast), then B as the appliance build. ? ? **RESOLVED: A first (done, §7), then B (B-P2, the SPA ingest/progress screen).**
docs/DRY_RUN_CUJ.md:43:- Path A's "point at the bucket" is a **CLI pre-step**, not in-UI ? full in-UI ingest is Path B.
docs/DRY_RUN_CUJ.md:46:## 6. Definition of done (the dry run)
docs/DRY_RUN_CUJ.md:47:- [x] A bucket video ? (engine processes) ? the **SPA loads its rallies** ? select ? compile ? **reel downloaded to the local machine**, all driven from the UI (Path A allows a one-line CLI pre-process). ? ? **Path A done (local, 2026-06-21)**, see §7.
docs/DRY_RUN_CUJ.md:49:- [x] *(Path B = B-P2, SPA-only)* the bucket video is pointed-at **from the UI** (no CLI pre-step). ? ? **DONE (live e2e, 2026-06-21), see §9.** In the SPA: entered a `gs://` URI ? `POST /api/process` ? polled `/api/jobs` ? **12 rallies loaded** ? Build ? **reel downloaded (1080p, 76s)**, all from the UI. Surfaced + fixed an engine compile bug along the way (engine [#289](https://github.com/Khelsutra/badminton-highlight-indexer/pull/289)). **Screencast:** `gs://khelsutra-rally-corpus/promo/khelsutra-bucket-cuj-screencast.mp4`.
docs/DRY_RUN_CUJ.md:57:4. **Build ? `/api/compile` ? reel** (1920×1080, 76.5s) ? **downloaded to `~/Downloads/khelsutra-dryrun/highlights_local_en.mp4`** (50.9 MB).
docs/DRY_RUN_CUJ.md:58:5. **Repeated with the UI in Hindi** (`?lang=hi`): the page + rally cards + "??? ?????" + "???? ??? ????? ??" all in Hindi; reel compiled + **downloaded** (`highlights_local_hi.mp4`). ? multilingual workflow confirmed.
docs/DRY_RUN_CUJ.md:76:**Already tested** (so this can't silently regress): `tests/test_api_endpoints.py::test_process_nonlocal_uri_skips_exists_and_fetches`, `?::test_process_hosted_mode_rejects_nonlocal_uri` (the hosted-mode counterweight), `?::test_process_unknown_scheme_returns_400`, `?::test_cli_nonlocal_input_fetches_and_keeps_uri`. So **B-P1 needs zero new engine feature code** ? the only Path-B work is **B-P2 (the SPA ingest/progress screen)**. *(The hermetic test fakes the network download via `m.storage.fetch`, so it proves the CODE routing, not the live GCS fetch ? that needs `google-cloud-storage` + ADC on the box, verified at deploy time / option 3.)*
docs/DRY_RUN_CUJ.md:80:## 9. ? Run result ? Path B, live in-UI bucket?reel CUJ (2026-06-21)
docs/DRY_RUN_CUJ.md:81:The full Path-B flow, driven **entirely from the UI** (no CLI pre-step), against a real local engine (`python -m backend.main --server`, Gemini detector, `GOOGLE_APPLICATION_CREDENTIALS` for the GCS fetch):
docs/DRY_RUN_CUJ.md:84:3. **Build reel** ? `/api/compile` ? **reel preview + download** ? MP4 saved locally (**1920×1080, 76s**).
docs/DRY_RUN_CUJ.md:92:- `2026-06-21` ? **§8 correction:** verified B-P1
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(`gs://`?`/api/process`?API-DB `video_id`) is already shipped + tested (M2/#280); §1/§2/§4/§6 reframed so Path B = SPA-only (B-P2). Sibling stale claims fixed in `LOCAL_APPLIANCE_ARCHITECTURE.md` (`/api/health` exists; bucket-URI ingest not a net-new endpoint; T12 engine-half done) + `README.md`.
docs/DRY_RUN_CUJ.md:93:- `2026-06-21` ? **§9: Path-B live e2e DONE** ? the in-UI `gs://`?reel CUJ ran end to end (12 rallies ? reel downloaded); surfaced + fixed the compile-from-bucket 500 (engine #289); screencast uploaded to the promo bucket. **DoD §6 box 3 ?.**

### 26. assistant (2026-06-21T13:57:37.185Z)

The khelsutra DRY_RUN_CUJ track is essentially complete (both DoD boxes ?, Path A + Path B live e2e done as of #57). This contradicts the engine DOC_STATUS §3a treatment of the cross-repo rows, and the khelsutra `docs/README.md` still calls DRY_RUN_CUJ the "? active goal (owner directive 2026-06-21)" without noting it's now DONE.

Let me verify the FRIENDLY_AND_MULTILINGUAL tracker state (DOC_STATUS engine row 49 says I18N_PLAN "in implementation"; khelsutra commits show SPA fully trilingual done) and confirm ANNOTATE_AND_TRAIN isn't indexed in khelsutra README.

### 27. user (2026-06-21T13:57:39.387Z)

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=== FRIENDLY_AND_MULTILINGUAL tracker rows (B/W series) ===
docs/FRIENDLY_AND_MULTILINGUAL.md:125:Legend: ? Todo · ? In progress · ? Done · ? Blocked/gated.
**One small PR per row**, branched from `origin/master`, no stacking.
docs/FRIENDLY_AND_MULTILINGUAL.md:129:| P0 | This tracked project doc + inbound pointers | ? | ?
| No | ? | [#41](https://github.com/avidullu/khelsutra/pull/41) · engine
[#275](https://github.com/Khelsutra/badminton-highlight-indexer/pull/275) |
docs/FRIENDLY_AND_MULTILINGUAL.md:130:| B1 | **SPA i18n scaffold + language switcher, en-only**
(react-i18next + detector + `en` catalog + switcher + persistence + **jsdom render tests** +
**browser-language negotiation tests** §6.5 N1/N3?N7) | B | P0 | No | ? |
[#42](https://github.com/avidullu/khelsutra/pull/42) |
docs/FRIENDLY_AND_MULTILINGUAL.md:131:| B2 | Extract SPA strings ? `en` catalog (~50 literals +
~16 derived; ICU plural keys); add **key-parity + no-bare-strings CI** | B | B1 | No | ? |
[#45](https://github.com/avidullu/khelsutra/pull/45) |
docs/FRIENDLY_AND_MULTILINGUAL.md:132:| A1 | Plain-language copy pass on the `en` catalog (de-
jargon; icon+text; non-color selection) | A | B2 | No | ? | ? |
docs/FRIENDLY_AND_MULTILINGUAL.md:133:| B3 | Tiny shared `t()` shim + `data-i18n` plumbing for
`site/`, en-only (proves shared catalogs across surfaces) | B | B2 | D-CATALOG-HOME | ? | ? |
docs/FRIENDLY_AND_MULTILINGUAL.md:134:| A2 | Localized **WhatsApp share** for the reel (SPA +
site) | A | B1/B3 | No | ? | ? |
docs/FRIENDLY_AND_MULTILINGUAL.md:135:| A6 | Plain-language framing for Request-Access
GPU/storage questions | A | B3 | No | ? | ? |
docs/FRIENDLY_AND_MULTILINGUAL.md:136:| A7 | Fix English-source recording gaps (add a placement
diagram) before localizing | A | ? | No | ? | ? |
docs/FRIENDLY_AND_MULTILINGUAL.md:137:| B4 | **Noto Sans Kannada + Devanagari** bundle (OFL
carve-out) ? *SPA done; reel watermark `font_path` = engine follow-up* | B | ? | D-FONT ? | ? |
[#46](https://github.com/avidullu/khelsutra/pull/46) |
docs/FRIENDLY_AND_MULTILINGUAL.md:138:| B5 | **Kannada (`kn`) catalog** ? render ? *SPA done
(AI-draft, native review pending); `site/` rides B3* | B | B2/B4 | D-LOCALES ? | ? |
[#47](https://github.com/avidullu/khelsutra/pull/47) |
docs/FRIENDLY_AND_MULTILINGUAL.md:140:| A4 | Mobile-first touch targets + tooltip-text ? visible
(SPA) | A | B1 | No | ? | ? |
docs/FRIENDLY_AND_MULTILINGUAL.md:141:| A5 | Friendly engine-error mapping over raw FastAPI
`detail` (SPA, client-side map) | A | B2 | No | ? | ? |
docs/FRIENDLY_AND_MULTILINGUAL.md:142:| B6 | **Hindi (`hi`) catalog** ? render (the
extensibility proof ? just a catalog) ? *SPA done (AI-draft, native review pending)* | B | B5 |
D-LOCALES ? | ? | [#48](https://github.com/avidullu/khelsutra/pull/48) |
docs/FRIENDLY_AND_MULTILINGUAL.md:143:| A8 | Background-processing reassurance + verdict/status
as icon chips (SPA) | A | B2 | No | ? | ? |
docs/FRIENDLY_AND_MULTILINGUAL.md:144:| A9 | Tap-only filter controls alongside the English
query bar (SPA) | A | B2 | No | ? | ? |
docs/FRIENDLY_AND_MULTILINGUAL.md:145:| B7 | *(deferred, cross-repo)* Engine **P2 backend
customer-facing localization** (report `customer_*` + rejection reasons) | B | engine |
**D-BACKEND-DEPTH, owner-gated** | ? | ? |
docs/FRIENDLY_AND_MULTILINGUAL.md:147:**? MILESTONE ? SPA fully trilingual (2026-06-21):** the
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`web/` Studio SPA now speaks **en + kn + hi** end to end (B1?B2?B4?B5?B6, all merged). A **Kannada-default browser auto-renders Kannada**, and the switcher walks kn ? hi ? en ? **proven by a Playwright screencast** with `locale: kn-IN` (`khelsutra-i18n-screencast.mp4`; frames captured per locale + `<html lang>` tracked). This meets the owner's success criterion **on the SPA surface**. **Still open for the full §9 DoD:** **B3** (the `site/` marketing surface via the shared `t()` shim ? the homepage/capture pages are still English-only), **A3** (recording-education in kn/hi), the **non-tech UX A-items**, and a **native-speaker review** of the AI-drafted kn/hi catalogs before a wide public launch. Do **not** archive until both surfaces are trilingual.
docs/FRIENDLY_AND_MULTILINGUAL.md:165:- [ ] §7 all rows ? (or explicitly deferred with a reason) ? flip **Status ? DONE** + completion note ? run engine `DOC_STATUS.md §2` archival checklist ? `git mv` to `docs/archives/`.
docs/FRIENDLY_AND_MULTILINGUAL.md:174:- `2026-06-21` ? owner locked **D6 (en+kn?hi)** + **D7 (OFL Noto carve-out)** and adopted **D8 (SPA-first)** + **D9 (catalogs in khelsutra)**; only D-BACKEND-DEPTH remains open. **B1 greenlit ? ? in progress.**

### 28. assistant (2026-06-21T13:57:44.045Z)

I now have a comprehensive picture. Let me do one final check on the engine local working-tree branch divergence (I'm on `feat/gcp-ops-agent-and-dryrun-guide`, 1 commit ahead, 0 behind origin/master per the earlier rev-list) and confirm the engine's NEXT_STEPS staleness relative to the merged M5/M6/M8. I have what I need; let me note the engine local branch is the Ops Agent work (NEXT_STEPS P1 item #6).

Now I'll compile the structured output.

### 29. user (2026-06-21T13:58:33.690Z)

Structured output provided successfully

### 30. assistant (2026-06-21T13:58:40.877Z)

I've mapped the cross-cutting registers across both repos and returned the structured findings. Key takeaways:

- **The central register is engine `docs/DOC_STATUS.md`** (LIVING, last full review 2026-06-21), but **origin/master has moved ahead of it** ? the serving track has merged M5/M6/M8 + bucket-ingest fixes (#286/#287/#288/#284/#289) that the register and `NEXT_STEPS.md` don't yet reflect. The project's own §7 tracker is current; the cross-cutting docs lag it.
- **Active/owner-gated tracks:** COMPUTE_DECOUPLED_SERVING (headline), the rally-quality/owned-model R-series cluster (blocked on golden-corpus growth + GPU features), PER_VIDEO_WORKSPACE, EXECUTION_POLICY, plus the cross-repo UX+localization track in khelsutra.
- **Staleness flags raised** (not fixed): DOC_STATUS/NEXT_STEPS serving rows, khelsutra README still calling the now-done DRY_RUN_CUJ the "active goal," and `ANNOTATE_AND_TRAIN.md` missing from the khelsutra index.